FLOATING SOLAR POWER PROJECT BY AMP ENERGY IN MADHYA PRADESH

Document Prepared by LGAI Technological Center S.A.
(Applus+ Certification)

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Project title	Floating Solar Power Project by AMP Energy In Madhya Pradesh
Project ID	VCS 4785[1]
Crediting period	06-June-2024 to 05-June-2031 (Inclusive both days)
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VCS Standard Version	Version 4.7

[1] Verra Search Page

Client	AMP Energy Markets India Private Limited
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Summary:

Validation purpose: The main purpose of this project activity is to reduce GHG emission by installing floating solar PV plant (the Project) at Omkareshwar water reservoir located near Punasa Tehsil, Khandwa District, Madhya Pradesh State, INDIA. Using the reservoir's area to produce clean energy and provide it to M.P. Power Management Company Limited (MPPMCL) is the aim of this project. The project activity is floating power project with total capacity of 140 MWp or 100 MW_{ac}, by **AMP Energy Green Seven Private Limited (Legal Owner)** a wholly owned subsidiary of **M/s AMP Solar India Private Limited (the Promoter)** is a special purpose vehicle incorporated by **AMP Energy Green Private Limited (Parent Company)** to develop a 140 MWp (100 MWAC) floating solar PV plant (the Project) at Omkareshwar water reservoir located near Punasa Tehsil, Khandwa District, Madhya Pradesh State, INDIA. **Amp Energy Markets India Private Limited (Project Proponent)** authorized by AMP Energy Green Seven Private Limited (Legal Owner) to act as a project proponent[3] for the project activity.

The project aims to displace electricity produced by fossil fuel power plants harnessing solar energy, Thus, the project reduced greenhouse gas emission and thereby contributes to sustainable development. The project being

[2] Team member was part of assessment team till 30-September-2024 only.

[3] In accordance with VCS Listing Representation, available on VCS project portal (https://registry.verra.org/app/projectDetail/VCS/4785)

a renewable energy generation activity, leads to reduction of greenhouse gas (GHG) emission that are real, measurable, and verifiable and also plays beneficial role in the mitigation of climate change.

The details of project capacity and location details for all the project activity are as follows:

S.I No.	Project Proponent	Legal Owner	Project Location	Installed Capacity MW_P	Installed Capacity MW_{AC}	Date of Commissioning
1.	AMP Energy Markets India Private Limited	AMP Energy Green Seven Private Limited	Ghogalgaon village, Punasa Tehsil, Khandwa District, Madhya Pradesh State	140	100	06-June-2024

During the 7 years of first crediting period i.e., 06-June-2024 to 05-June-2031, the project will replace anthropogenic emissions of greenhouse gases (GHG's) estimated to be approximately 197,846 tCO_2e per year, thereon total GHG emission reductions for the chosen 07-year crediting period will be 1,384,927 tonnes of CO_2e.

The project proponent has applied the baseline and monitoring methodology ACM0002: Grid-connected electricity generation from renewable sources - Version 22.0.

The objective of this validation activity is to have an independent third-party for the assessment of the project design, estimated ER sheet and to ensure a thorough assessment of the proposed project activity against the applicable VCS requirements. In particular;

- The project's monitoring plan is assessed against "ACM0002: Grid-connected electricity generation from renewable sources - Version 22.0.
- VCS Validation and Verification Manual v3.2.
- VCS Standard v4.7.
- VCS Program Guide v4.4.

Validation is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of estimated verified carbon units (VCUs).

A risk-based approach has been followed to perform this Validation activity. In the course of Validation, 02 Corrective Action requests (CARs) and 01 Clarification Requests (CLs) were raised and successfully closed and 00 Forward action request (FARs) were raised during the Validation. The review of the project description and additional documents related to baseline and monitoring methodology; the subsequent background investigation, follow-up interviews and project owners have provided LGAI Technological Center S.A. (Applus+ Certification) (hereinafter may be referred to as the VVB) with sufficient evidence to validate the fulfilment of the stated criteria of VCS.

Validation conclusion: - The conclusion of this report shows, that the project, as it was described in the project documentation, is in line with all criteria applicable for the validation. This validation is based on the information made available to LGAI Technological Center S.A. (Applus+ Certification) and the engagement conditions are detailed in this report. VVB's team didn't identify any restriction or uncertainties related to validation of the project activity.

CONTENTS

1 INTRODUCTION

1.1 Objective

LGAI Technological Center S.A. (hereinafter referred to as Applus + Certification or the VVB) has been appointed by "AMP Energy Markets India Private Limited" to perform the Validation of the project titled "**Floating Solar Power Project by AMP Energy In Madhya Pradesh**" under VCS Standard v4.7 and VCS Program Guide v4.4. The objective of this Validation activity is to have an independent third-party for the assessment of the Project documentation, ER sheet(s) and to ensure a thorough assessment of the proposed project activity against the applicable VCS requirements. In particular;

- The project's baseline is assessed against "ACM0002: Grid-connected electricity generation from renewable sources - Version 22.0.
- The project's monitoring plan is assessed against "ACM0002: Grid-connected electricity generation from renewable sources - Version 22.0.
- VCS Validation and Verification Manual v3.2.
- VCS standard v4.7.
- VCS Program Guide v4.4
- VCS Registration and Issuance Process version 4.4

LGAI Technological Center, S.A. (Applus+ Certification) as the Validation/Verification Body (VVB) for the project activity is accredited as a DOE by UNFCCC and also meets the competence requirements as set out in the normative reference ISO 14065:2020.

Validation is a requirement for all VCS projects and is seen as necessary to, through the provision of objective evidence, provide assurance to stakeholders of the quality of the project and that the requirements for a specific intended future use or application have been fulfilled regarding the GHG claims (*i.e.* generation of estimated Verified Carbon Units (VCUs)).

1.2 Scope and Criteria

The scope of the Validation is the independent and objective review of the Project Description. The VCS PD is reviewed against the relevant criteria (See Section 1.1. above) including the approved baseline and monitoring methodology (ACM0002: Grid-connected electricity generation from renewable sources - Version 22.0).

The validation has been conducted in accordance with the VCS Standard version 4.7, other valid and applicable VCS Regulatory Documents and templates (see Appendix 2 of this Validation Report below), UNFCCC rules and requirements (where applicable) and associated decisions of the VCS Secretariat and the UNFCCC Executive Board (where applicable) (hereinafter also can be referred to as regulatory documents).

The assessment team has employed a risk-based approach to assess the completeness and accuracy of the claims and conservativeness of the assumptions in the VCS PD. The main focus of the assessment team is to identify the significant risks for the project implementation and the generation of VCUs. The validation is not meant to provide any consulting towards the project proponent However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project design.

The only purpose of the validation is its usage during the registration/issuance processes as part of the VCS project cycle. Therefore, LGAI Technological Center S.A. (Applus+ Certification) can't be held liable by any party for decisions made or not made based on the validation opinion, which will go beyond that purpose.

1.3 Reasonableness of Assumptions

Applus+ Certification validation approach is based on the understanding of the risks associated with estimation of GHG emission reductions data and the controls in place to mitigate these.

Applus+ Certification planned and performed the Validation by obtaining objective evidence that substantiate the reasonableness of PP's assumptions and determine whether the project and its GHG statement conforms with the regulatory documents, as well as evaluates the limitations and methods that support a claim about the outcome of future activities, in accordance with the VCS Standard 4.7. Section 4.1.2 (1).

Likewise, Applus+ Certification collected as well other information and explanations that it considers necessary to give reasonable assurance that reported estimated GHG emission reductions and the assumptions that substantiate these estimations are fairly stated.

In our opinion, the estimated GHG emissions reductions were calculated correctly on the basis of the approved baseline and monitoring methodology ACM0002: Grid-connected electricity generation from renewable sources - Version 22.0 and the VCS Standard, v4.7.

1.4 Summary Description of the Project

The main purpose of this project activity is to reduce GHG emission by installing Floating Solar Photovoltaic (SPV) Plant in Omkareshwar water reservoir located near Punasa Tehsil, Khandwa District, Madhya Pradesh State states of India by AMP Energy Green Seven Private Limited. AMP Energy Green Seven Private Limited has developed, installed, financed, operates, and owns the floating solar power plant and use the generated electricity supplied to national (Indian) grid.

AMP Energy Markets India Private Limited is looking forward to earn carbon credits under VCS mechanism for this Floating solar plant. The details of project capacity and location details for all the project instances are as follows:

S.I No.	Project Proponent	Installed Capacity (MW)	Latitude (N)	Longitude (E)
1.	AMP Energy Markets India Private Limited	140 MW_P/100 MW_{AC}	22°11'33.0" N (22.1925)	76°13'04.0" E (76.2177)

During the 7 years of first crediting period i.e., 06-June-2024 to 05-June-2031 (inclusive both days), 211,556 MWh/year electricity generated and 1,480,892, amount of electricity generated during crediting period, the project will replace anthropogenic emissions of greenhouse gases (GHG's) estimated to be approximately 197,846 tCO_2e per year, thereon total GHG emission reductions for the chosen 07-year crediting period will be 1,384,927 tonnes of CO_2e.

2 VALIDATION PROCESS

2.1 Method and Criteria

The Validation process has been conducted following the regulatory documents and based on the ISO 14064-3:2019 and ISO 14065:2020 using standard auditing techniques as required by the VCS Standard version 4.7.

The Validation process has been conducted through an onsite inspection (see below), thus, no need for a risk analysis for remote inspections has been determined for this Validation (*cf.* VCS Standard v4.7. Para 4.1.13).

No ICT means have been used for the conduction of this Validation process other than those for the provision of documents and transferring information and undertake normal communications with the PP(s). Those ICT means supporting the process in this sense, have been used to an extent sufficient to cover the background necessities of the Validation process and have demonstrated enough effectiveness to conduct it in accordance with the relevant applicable regulatory requirements. No virtual sites have been included in the Validation process according to the definition in the IAF MD4:2023 document. Instead, the documentation has been provided upon request from the VVB by other means (such as physical documents, mail, transfer applications, etc.).

Before the assessment begins, members of the team covering the technical scope(s), sectoral scope(s), and relevant host country experience for evaluating the VCS project activity are appointed in accordance with the Applus+ Certification's procedures and policies, that are in accordance with the ISO 14065:2020 regulatory provisions.

Applus+ Certification has completed a strategic review and risk assessment of the project's activities and processes in order to gain a full understanding of:

- Activities associated with all the sources contributing to the project's emissions and emission reductions, including leakage if relevant;
- Protocols used to estimate or measure GHG emissions from these sources;
- Collection and handling of data;
- Controls on the collection and handling of data;
- Means of validating the estimated data; and
- Compilation of the information in the VCS PD.

Based on the above, the VVB prepared a Validation Plan and applied an evidence-gathering plan designed to collect sufficient and appropriate evidence upon which to base the conclusion and to consider the inherent risk for the validation. The evidence-gathering plan is based on analytical procedures and tests and considers the thresholds for materiality as per the regulatory documents applied for this Validation (5% as per the applied para 4.1.10(4) of VCS Standard version 4.7).

The applied evidence-gathering plan is used by the VVB to determine the conformity of the statements made by the client against the principles and requirements of the applied regulatory documents during this validation.

The evidence-gathering plan designed by the VVB takes into consideration, as a minimum:

- The selection and management of the data and information;
- The processes for collecting, processing, consolidating, aggregating and reporting the data and information;
- Systems and processes in place to ensure the validity and accuracy of the data and information;
- The design and maintenance of the system to control the data and information;
- Systems, processes and personnel that support system in place, including activities for ensuring data quality;
- The plans for instrument maintenance and calibration where appropriate;

Applus+ Certification has validated the design of the monitoring plan and the data presented in the Project Description version 04.0, dated on 16-October-2024 for the crediting period applicable to this validation process.

The process of validation involved the following relevant milestones:

- The signature of the contract for validation between the PP and the VVB;
- A kick-off meeting/communication to require the documentary material to perform an initial desk review;
- An initial desk review of the documentary evidences prepared by the Project Proponent;
- The issuance of findings coming from this initial desk review (if any);
- The conduction of an onsite visit for the Project locations (No Sampling approach has been performed), including interviews with project proponent and project stakeholders (see Sections 2.3 and 2.4 below);
- The issuance of findings from the On-site audit and communication of such deviations (if any) to the Project Proponent;
- The assessment of their resolution and closure, when appropriate;
- Once all the findings have been closed, the elaboration of this Validation Report, in which such findings (if any) are described;
- The conduction of an independent Technical Review;
- Upon conclusion of the Technical Review, the VVB may perform a final quality check before the issuance of the Final Validation Report (FVR) and signature of the corresponding Validation Deed.

The process of this validation assessment has not implied any sampling plan.

Appointment of the assessment team: -

According to the sectoral scope / technical area and experience in the sectoral or national business environment, LGAI Technological Center S.A. (Applus+ Certification) has composed a project assessment team in accordance with the appointment rules in the internal Quality Management System of LGAI Technological Center S.A. (Applus+ Certification).

The composition of audit team is approved by the LGAI Technological Center S.A. (Applus+ Certification) Central Site during its Contract Review stage, ensuring that the required skills are covered by the team.

The qualification levels for Assessment Team members that are assigned by aforementioned appointment rules are as presented below:

- Lead Auditor (LA).
- Auditor (A) /Auditor in Training (AiT).
- Technical Expert (TE).

- Technical Reviewer (TR).
- Other supporting roles such as Country Expert (CE) or Financial Expert (FE).
- Any of the above-mentioned roles in training (iT, e.g. AiT for auditor in training/TeiT for Technical Expert in Trainee).

The Sectoral Scope / Technical Area required knowledge linked to the applied methodology(ies) is covered by the Assessment Team as shown below:

Name	Role	SS/ TA Coverage	Financial aspect	Host country experience	Attendance to on-site visit
Mr. Ishan Shrivastava	Lead Auditor/Technical Expert	YES (1.2)	YES	YES	YES
Dr. Atul Takarkhede	Auditor/Financial Expert	YES (1.2)	YES	YES	NO
Mr. Khagesh Sharma	Auditor-in-Training	NO (1.2)	NO	YES	YES
Ms. Anjali Singh	Auditor-in-Training	YES	NO	YES	YES
Mr. Sarthak Jain	Observer	NO	NO	YES	YES
Mr. Ravi Jangid[4]	Observer	NO	NO	YES	YES
Dr. N. Premjit Singh	Technical Reviewer	YES (1.2)	YES	N/A	N/A

The complete list of CVs is included as Appendix 4 of this report.

Document Review: -

The project description (VCS PD) submitted by the Client was reviewed against the approved methodology and other relevant criteria to verify the correctness, credibility, and interpretation of the presented information. Furthermore, a cross-check between information provided and information from other sources has been done. A complete list of all documents and evidence material reviewed is included in this report below in appendix 2

Onsite visit:

An on-site audit on 19-March-2024 is conducted by LGAI Technological Center S.A. (Applus+ Certification) assessment team who performed interviews and on-site inspection with project's stakeholders to confirm selected information and to resolve issues identified in the document review. No Sampling approach has been performed during the visit; Assessment team visited

[4] Team member was part of assessment team till 30-September-2024

all site during the above-mentioned period. The detail is provided in this report in the below sections Resolution of Clarification and Correction Action Request.

Resolution of Clarification and Correction Action Request: -

The objective of this phase of the validation was to resolve the requests for corrective actions and clarification and any other outstanding issues which need to be clarified for LGAI Technological Center S.A. (Applus+ Certification) positive conclusion on the project design. The Corrective Action Requests and Clarification Requests raised by LGAI Technological Center S.A. (Applus+ Certification) were resolved during communications between the Client and Applus+ Certification to guarantee the transparency of the validation process, the concerns raised and responses given are summarized below in the appendix 3

The final version of VCS PD submitted by project proponent serves as the basis for the final assessment presented. Additional changes to the project during the validation is not considered to be significant with respect to the main VCS objectives. The two VCS main objectives are the reduction of anthropogenic GHG emissions and the contribution of sustainable development to the host country.

Internal quality Control: -

As final step of a validation the final documentation including the validation report and the protocol have to undergo an internal quality control by the technical review committee. Once approved by the Technical Review Team, each report has to be finally approved by the authorized personnel of the VVB (either the Technical Manager or other authority in case the Technical Manager is part of the Validation and/or Verification-Certification (VVC) Team for the project).

After confirmation of the PP, the validation opinion and relevant documents are submitted to the VCS Registry.

2.2 Document Review

The details of the document observed during the Validation process are listed below in appendix 2 of this report.

2.3 Interviews

Organization	Name of Persons	Designation	Topics discussed	Team Member
AMP Energy Green Seven Private Limited	Mr. Shriprakash Rai	PP's Representative	Project Implementation, O&M activities, LSC etc, and	Mr. Ishan Shrivastava (Lead Auditor & Technical Expert)
	Mr. Piyush Jain,	Manager		

Organization	Name of Persons	Designation	Topics discussed	Team Member
AMP Energy Green Seven Private Limited	Mr. Uma Shankar	AMP Site In-charge	technical specifications	& Mr. Khagesh Sharma (Technical Expert in Training) & Ms. Anjali Singh (Auditor-in-Training) & Mr. Sarthak Jain and Mr. Ravi Jangid (Observer)
	Mr. Saurabh Pandey	AMP Engineer		
	Mr. Md. Sheerm			
	Mr. Vishal Souner	Site Supervisor		
	Mr. Sachin Chouhan	Store Incharge		
Enking International	Mr. Rutuparn Nimonkar	All Project sites	ER calculations, VCS PD, Project description, eligibility, baseline, additionality, IRR calculations etc.	
Local stakeholders	Mr. Mohan kumar	Local stakeholder	Local Stakeholder consultation, ongoing communication	
	Mr. Kapil Gour	Local stakeholder		
	Mr. Ankit patel	Local stakeholder		
	Mr. Umakant meena	Local stakeholder		

2.4 Site Visits

As a part of validation an onsite visit has been performed by VVB team. Physical site visit for project site was conducted for the project activity on 19-March-2024. During the onsite visit representative of the project proponent, operation and maintenance staff were interviewed; personal responsible for monitoring, collecting data and management, and QA/QC procedure. No Sampling approach has been performed. The details of the personal interviewed are mentioned in the table below.

Duration of Onsite Visit: 19-March-2024				
Sr. No.	Activity performed on-site	Site location	Date	Team Member

1.	Assessment team checked the implementation of the project, Baseline emission, Emission reduction calculation, technical description of the project and Monitoring.	Omkareshwar water reservoir located near Punasa Tehsil, Khandwa District, Madhya Pradesh State states of India	19-March-2024	**Mr. Ishan Shrivastava** (Lead Auditor & Technical Expert) **Mr. Khagesh Sharma** (TEiT), **Ms. Anjali Singh** (AiT) & **Mr. Sarthak Jain and Mr. Ravi Jangid** (OBS)

2.5 Resolution of Findings

The objective of this phase of the validation was to resolve the requests for corrective actions and clarification and any other outstanding issues which need to be clarified for LGAI Technological Center S.A. (Applus+ Certification)'s positive conclusion on the project design. The Corrective Action Requests and Clarification Requests raised by LGAI Technological Center S.A. (Applus+ Certification) were resolved during communications between the Client and LGAI Technological Center S.A. (Applus+ Certification) to guarantee the transparency of the validation process, the concerns raised and responses given are summarized below in the Appendix 3.

The final VCS PD submitted by project proponent serves as the basis for the final assessment presented. Additional changes to the project during the validation process is not considered to be significant with respect to the main VCS objectives. The two VCS main objectives are the reduction of anthropogenic GHG emissions and the contribution of sustainable development to the host country.

Areas of validation findings	No. of CL	No. of CAR	No. of FAR
Project design document and Monitoring report	00	00	00
Description of project activity	CL 01	CAR 01	00
Application of selected baseline and monitoring methodology and selected			
- Applicability of methodology and standardized baseline	00	00	00
- Deviation from methodology	00	00	00
- Clarification on applicability of methodology, tool and/or standardized baseline	00	CAR 02	00
- Demonstration of additionality	CL 01	CAR 02	00

Areas of validation findings	No. of CL	No. of CAR	No. of FAR
- Emission reductions	00	00	00
- Monitoring plan	00	00	00
- Stakeholder's consultation process	CL 01	00	00
- Public comments	00	00	00
Others (please specify)-Matter related to double counting- for validation	CL 01	00	00
Others (please specify)- Site visit	00	00	00
ER achieved – Actual ER achieved calculations, Compliance of the registered monitoring plan with the methodologies including applicable tools and standardized baselines Compliance of monitoring activities with the registered monitoring plan Compliance with the calibration frequency requirements for measuring instruments Assessment of data and calculation of emission reductions or net removals Assessment of reported sustainable development co-benefits	00	CAR 02	0
Total (Validation)	01	02	00

The list of findings and their resolution is presented in Appendix 3 of this report.

2.5.1 Forward Action Requests

No FAR was raised during this validation process. Please refer Appendix 3 for details.

3 VALIDATION FINDINGS

3.1 Project Details

The project activity is floating solar power project activity installed in the proposed 140 MWp (100 MW_{ac}) floating solar PV plant is developed on the surface of Omkareshwar water reservoir located near Punasa Tehsil of Khandawa district of Madhya Pradesh State of India. Total capacity of the project activity is 140 MW_p/100MW_{ac}. Moreover, VVB team had conducted the on-site visit and it is verified that; the floating solar PV plant's electricity has been transferred to the utility substation at Saktapur's 220 kV substation. The electricity generated from project activity (operational solar power plant) is connected to the National grid.

The project activity is already commissioned as mentioned in section 1.9 of the project description (VCS PD). This is checked and confirmed from the commissioning certificate[/01/] of the project activity, also verified during the site visit.

The project activity is a green field project which results in an estimated GHG emission reduction of 1,384,927 tCO_2e over 7 year of renewable crediting period, and 197,846 tCO_2e in a year.

Technology: -

The floating solar power plant has the total capacity of 140 MW_p/100MW_{ac}, having solar panels & inverters of different capacity and a control cabinet. This is a large-scale solar grid-connected electricity generation project activity which supplies electricity to the different consumer.

In the baseline scenario the main source of emission was found to be CO2 as electricity was generated mainly through fossil-fuel based power plants whereas in project scenario the electricity is generated by the Floating Solar Power plant thereby reducing the CO2 emissions.. Thus, this project will reduce GHGs emissions and generate GHGs reductions throughout the project.

The project utilizes solar photovoltaic to convert energy from the sun into electricity for generation power, and inverter is installed to change the direct current generated from solar photovoltaic to alternating current. Then connecting the solar panel with investors and transmitting it to the end consumer.

The solar power plant consists of main equipment as follows:

- Solar PV Modules
- Central inverters
- Module Mounting system
- Grid connect equipment
- Monitoring system
- SCADA
- Cables & connectors

VVB team further verified that, all equipment specifics used in the project activity were confirmed from the manufacturer's specifications, and the information was verified to be compatible with the project description. details of the all the equipment used in the project activity is found consistent with the details given in the project description. The details of the solar module and invertors is given in below table: -

Description	Specification
Type of PV Module	Mono C-Si Module
Power rating (Wp)	540 Wp or higher
Module Power Rating Tolerance (%)	+Ve
System Voltage (DC)	1500 V
Temp. Coefficient of power (%/°C)	- 0.34%/Degree C
Number of cells	259,280
Module Make	Longi
Series fuse rating(A)	30A

Glass	Dual Glass, 2+2 heat strengthened glass
Module efficiency %	21.5%

Inverter –

Input DC	Value
Max. PV input voltage	1500 V
Min. PV input voltage / Start-up input voltage	960 v/ 990 V
MPP voltage range for minimal power	960 , 1300 V
No. of independent MPP inputs	1
No. of DC inputs	30
Max. PV input current	6112 A
Output AC	
AC output power	5750 KVA @ 25 degrees Celsius
Max. AC output current	5030 A
Nominal AC voltage	660 V
AC voltage range	561 -726 V
Nominal grid frequency / Grid frequency range	50 Hz / 45 -55 Hz, 60 Hz / 55 -65 Hz
THD	<3 %
DC current injection	<0.5 %
Power factor at nominal power / Adjustable power factor	>0.99 /0.8 leading
Efficiency	
Max. efficiency	99.0 %
European efficiency	98.7 %
Protection	
DC input protection	Load break switch + fuse
AC output protection	Circuit breaker
Overvoltage protection	DC Type I + II / AC Type II
Grid monitoring / Ground fault monitoring	Yes / Yes
Insulation monitoring	Yes
Overheat protection	Yes
General Data	

Dimensions (W*H*D)	5020 * 2400 * 1100 mm
Weight	6 T
Isolation method	Transformer less
Degree of protection	IP65

- **Power Station and Transformer:** - The project comprises four electrical control buildings that oversee the conversion of direct current into alternating current. Specifically, the inverters are the off-grid Photovoltaic Inverters & each inverter boasting a nominal AC output power capacity.
- **Control Building:** - The control building has a performance display for the solar panels and is responsible for recording their operation. Additionally, a CCTV system is in place to ensure safety in and around the project.

CL 01 & CAR 01 were raised on this section and closed successfully. Please refer Appendix 3 for further details.

Hence, Assessment team confirms that the purpose of this project activity is to reduce GHG emission by installing Solar Photovoltaic on rooftop solar power plant. Same is confirmed through document review and onsite visit with the project site personnel. Final VCS PD also reflects the same. Thus, found acceptable as VCS PD is accurate and provided complete project information.

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Audit history	Validation for the 1st crediting period 06-June-2024 to 05-June-2031 (Inclusive both days)
Sectoral scope	1.2 Energy Industries (renewable/non-renewable)
AFOLU project category, if applicable	Assessment team verified that; project activity is group captive solar power project thus this criterion is not applicable.
Project activity type	Renewable Energy Projects
General eligibility of the project to participate in the VCS Program	• VVB team verified that, project activity is floating solar power plant and it is grid connected project moreover it is consistent and not excluded under Table 1 in section 2.1.3 of the VCS Standard v4.7. • This project activity is non-AFOLU project and completing validation within two years of the project start date as per clause 3.8.1 of VCS Standards 4.7.

Item	Evidence gathering activities, evidence checked, and assessment conclusion
	• The opening meeting with the VVB took place only after the project has been placed under validation in the VCS Registry. During desk review, VVB team found that project proponent submitted project description and accomplice document to VERRA for project pipeline listing afterwards project was open for public comments for the period 13-Feburary-2024 to 14-March-2024. Further, VVB team assessed that the earliest commissioning date of the project instance is 06-June-2024, which is in line with the commissioning certificate of the project activity. Accordingly, the validation deadline for the project activity will be 05-June-2026 (as per section 3.8.1 of VCS standard version 4.7). VVB team verified that, generated electricity from the grid is supplied to the grid and project proponent has applied the ACM0002 and the applied methodology is eligible under the VCS Program. The applicability condition of the applied methodology is found consistent in section 3.2.2 of the validation report, including capacity limits and scale. • The project is not a fragmented part of a larger project or activity that would otherwise exceed such limits.
AFOLU project eligibility, if applicable	Not applicable
Transfer project eligibility, if applicable	Not applicable as the project is not being transferred from any other approved GHG Program.
Project design	As per para 3.6 of the VCS Standards version 4.7, the proposed project activity is a single installation.
Project ownership	The renewable energy technology installed at site of the project activities as part of the proposed project activity are under ownership of respective SPV/Project Proponent. AMP Energy Markets India Private Limited is Project Proponent of floating solar power plant. The commissioning certificate for project activity are the supporting documents to demonstrate the project ownership. This demonstrates the right of use according to clause 3.7.1 of VCS Standard Version 4.7 – "*a project ownership arising by*

Item	Evidence gathering activities, evidence checked, and assessment conclusion
	virtue of a statutory, property or contractual right in the plant, equipment or process that generates GHG emission reductions and/or removals”. Assessment team has checked the legal document which includes the commissioning certificate and Board Resolution for each instance to confirm the ownership of the project.
Project start date	06-June-2024 Assessment team has checked the generation details and further it has been crosschecked with invoices and it is found that the project has start its operation on 06-June-2024 and it is found consistent with para 3.8 of the VCS Standards version 4.7.
Project crediting period	06-June-2024 to 05-June-2031 (Inclusive both days) Assessment team verified that; project activity is non-AFOLU project and PP has chosen, crediting period 7years, renewable up to twice, it is find consistent with para. 3.9.2 of the VCS Standard Version 4.7 7 Year Crediting period (Renewable)
Project scale	Assessment team verified that; the annual emission reduction by the project activity is 197,846 tCO_2e . As per para 3.10.1 (1) VCS standards version 4.7, the emission reduction of project activity is less than 300,000 thus the scale of the project is Project.
Likelihood of achieving estimated GHG emission reduction or removals	During review, VVB team observed that the estimated electricity generation will be transmitted to 33/220 saktapur sub-station and will be supplied to Indian national grid through the long-term power purchase agreement signed between Consumer and investor in which annual generation is estimated through the technical life of solar panels i.e., 25 years. Also, PP submitted third party PLF report prepared by TUV SUD South Asia (India), and third-party detailed project report prepared by AMP Energy Green Seven Pvt. Ltd. representing maximum annual generation has as 228 GWh, whereas in ER sheet average for 7 years generation is calculated as 211,556. Thus,

<table>
<tr><th>Item</th><th>Evidence gathering activities, evidence checked, and assessment conclusion</th></tr>
<tr><td></td><td>VVB confirm chance of likelihood of achieving estimated GHG emission reduction or removals.</td></tr>
<tr><td>Technologies and measures implemented by the project activity or activities</td><td>AMP Energy Green Seven Pvt. Ltd. consists of the installation of Floating solar power plant with a capacity of 100 MW_{AC} (140 MW_{P}), located in Madhya Pradesh State of India. Technical specification details have been provided in section 3.1 of the validation report.</td></tr>
<tr><td>Implementation schedule of the project activity or activities</td><td>The project activity has been commissioned on the 06-June-2024, same has been verified through the commissioning certificate issued by Rewa Ultra Mega Solar Ltd.</td></tr>
<tr><td>Project location</td><td>Madhya Pradesh State of India.

Geo-coordinates: -
<table>
<tr><th>Sl.No.</th><th>Project Location</th><th>Latitude</th><th>Longitude</th></tr>
<tr><td>1</td><td>Village Ghogalgaon, Omkareshwar water reservoir located near Punasa Tehsil, Khandwa District, Madhya Pradesh State, INDIA</td><td>22°11'33.0" N (22.1925)</td><td>76°13'04.0" E (76.2177)</td></tr>
</table>
</td></tr>
<tr><td>Conditions prior to project initiation</td><td>Assessment team verified that; prior to the initiation of the project activity, the equivalent amount of electricity would have been drawn from grid connected or new power plants, in Indian Grid. The grid is predominantly coal based and therefore is a major source of carbon dioxide emissions in India.</td></tr>
<tr><td>Project compliance with applicable laws, statutes and other regulatory frameworks</td><td>Assessment team confirms that the project has received necessary approvals for development and commissioning for the proposed Solar PV project from the state Nodal agencies and is in compliance to the local laws and regulations. Assessment team checked the Commissioning certificates, Net metering agreements, Installation report for Solar Plants to confirm the project capacity and its relevant statutory requirements as per the host country regulations.</td></tr>
</table>

Item	Evidence gathering activities, evidence checked, and assessment conclusion
	Assessment team noted that the project fulfils the norms put down by Central Pollution Control Board norms. As per Central Pollution Control Board (Ministry of Environment & Forests, Govt. of India), final document on revised classification of Industrial Sectors under Red, Orange, Green and White Categories (29/02/2016). Being a renewable power project, it falls under the category of White and thus these projects do not need clearance for Consent to operate and only needs to inform the relative State Pollution Control Board. The same is done for the project and thus it can be confirmed that it follows the local laws of the host country. The relevant national laws and regulations pertaining to generation of energy in India are: • Electricity Act 2003[5] • National Electricity Policy 2005[6] • Tariff Policy 2006[7] • The Project activity conforms to all the applicable laws and regulations in India: • Power generation using solar energy is not a legal requirement or a mandatory option. • There are state and sectoral policies, framed primarily to encourage solar power projects. These policies have also been drafted realizing the extent of risks involved in the projects and to attract private investments. • The Indian Electricity Act, 2003 (May 2007 Amendment) does not influence the choice of fuel used for power generation. • There is no legal requirement on the choice of a particular technology for power generation The project complies with all relevant local and national law regulation of host country India.

[5] https://cercind.gov.in/Act-with-amendment.pdf

[6] https://indiatransmission.org/listdocuments/Policy/NEP?doc_id=1101

[7] https://cea.nic.in/wp-content/uploads/legal_affairs/2020/09/Tariff%20policy.pdf

Item	Evidence gathering activities, evidence checked, and assessment conclusion
	Further, VVB team assessed the regulatory surplus details mentioned under section 3.5.1 of VCS PD and confirmed that project activity is not mandated by any Host country's law.
Double counting and participation under other GHG programs	The project is seeking registration under VCS (with Project ID VCS 4785[8]) only. Assessment team further added that, project activity is not receiving or not seeking any credit for reductions and removals from a project activity under another GHG program i.e. CDM, GS4GG, GCC. Assessment team verified and has checked the other GHG mechanism web interface and it is confirmed that project activity is not register in other mechanism, as per para 3.23.1 of the VCS Standards version 4.7, project proponent has submitted the no double accounting declaration. Project has provided the information that; project is not register other than VCS and project activity is not rejected by other mechanism. Moreover, project proponent has submitted the no double accounting declaration/13/ which states that the net GHG emission reduction generated by the project activity will not be used for compliance with any other emission trading programme or to meet binding limit on GHG emission. Assessment team also checked independently other registries and found that project is not participating in GHG programs other than VCS program.
No double claiming with emissions trading programs or binding emission limits	The project activity is not participating in other emission trading programs or binding emission limits. The letter of undertaking has been furnished by the project proponent confirming that net GHG emission reduction or removal generated by the project will not be used for compliance with an emission trading programme or to meet the binding limit on GHG emission. There is no emissions trading programs or binding emission limits in India. CL 01 & CAR 01 were raised on this section and closed successfully. Please refer Appendix 3 for further details

[8] Verra Search Page

Item	Evidence gathering activities, evidence checked, and assessment conclusion
No double claiming with other forms of environmental credit	VVB team verified that, the project only seeking registration under the VCS mechanism with (VCS Ref No. 4785). The VVB team reviewed other mechanisms. The VVB team also confirmed that PP had submitted them a declaration of no double accounting stating that there was no double accounting and verified that project activity wasn't seeking double claims for environmental credits.
Supply chain (Scope 3) emissions double claiming	Assessment team verified that; Project is group captive solar project, with capacity of 140 MWp i.e. 100 MWac. Assessment team further verified that, In-line with para 3.24.7 of VCS Standard V4.7, the PP doesn't fall in the supply chain as the PP is not a buyer or seller of a product (a good or service) and whose product emission footprint is changed by the project activities specified in the VCS PD. Hence, these scope 3 emissions may be accounted for tenants but not to the PP. CL 01 was raised on this section and closed successfully. Please refer Appendix 3 for further details
Sustainable development contributions	Project activity is claiming SDG 7, SDG 8, & SDG 13 **SDG 7**: - VVB team verified that, this project is renewable solar power project and the installations started operation from 06-June-2024 and same was verified after reviewing the monthly generation report & same has been cross check with the monthly invoices and found it consistent. Estimated annual generation is 211,556 MWh/year. Though it is verified that; the generated electricity transmitted to the grid, same is confirmed through the onsite visit observations. The generated power from the project activity is the clean energy and continuously monitored by the energy meters installed at the site and included in the monitoring plan in the VCS project description. **SDG 8**: - VVB team verified that, Project proponent has operated the plant since 06-June-2024 and complies with targeted SDGs so far, Providing the employment to 22 person

Item	Evidence gathering activities, evidence checked, and assessment conclusion
	and 1 training. and same has been also verified during onsite visit. **SDG 13:** - VVB team found that, the project's positive impacts, which will prevent around 197,846 tCO_2 annually. The clean energy generated by the project activity is regularly measured by the energy meters that are installed at the site and are part of the PD's monitoring plan.
Additional information relevant to the project	• Assessment team verified that; the project activity is not AFOLU project thus it is not required. • **Commercially sensitive information:** - No commercially sensitive information has been excluded from the public version of the project description. • The project activity is not-grouped project, hence not applicable to the project activity. VVB team concludes that the proposed project activity is eligible to register under VERRA registry and all calculation related to emission reduction are found correct. Thus accepted.

3.2 Safeguards and Stakeholder Engagement

3.2.1 Stakeholder Engagement and Consultation

3.2.1.1 Stakeholder Identification

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Stakeholder identification	Project proponent has identified stakeholders on the basis of affected local communities within nearby involvement project radius, government institutions, General citizens, Schools & institution. The stakeholder consultation identification, stakeholder meeting details and outcomes is described in the detailed project report. Assessment team checked the local stakeholder records, meeting images and stakeholder identification found appropriate. CL 01 was raised on this section and closed successfully. Please refer Appendix 3 for further details.

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Legal or customary tenure/access rights	The assessment team conducted a site visit and reviewed the documents, confirming that the project activity involves floating solar projects. Before commissioning, the project proponent obtained all necessary consents from the state electricity authority. Since the project is installed in the rooftop of the beneficiaries it does not need land clearance for the installation of the project. Assessment team has checked the ownership documents and verified that, project is not violating any customary, legal and access right of the stakeholder. Same has been check during the interview with project proponent, site in-charge and stakeholder present there.
Stakeholder diversity and changes over time	Assessment team verified that, the project has no impact on the social, economic, and cultural diversity within stakeholder groups. Same has been cross verified during interview with stakeholder and project proponent.
Expected changes in well-being	The project generates job opportunities, both directly and indirectly, for the local community during its construction and operation. Additionally, the community will experience positive economic effects as local businesses see increased income from the project's financial contributions. Assessment team has interviewed the local stakeholder and information found correct.
Location of stakeholders	Assessment team verified that; there is no impact by the project activity to the location of stakeholders, Indigenous Peoples (IPs), local communities (LCs), customary rights holders, and areas outside the project area.
Location of resources	Assessment team verified that; the project activity is floating solar plants; thus, this section is not applicable as this project activity does not contain any location of territories and resources which stakeholders own or to which they have customary access.

3.2.1.2 Stakeholder Consultation and Ongoing Communication

Item	Evidence gathering activities, evidence checked, and assessment conclusion	
Stakeholder engagement process	**LSC Location**	**Meeting Date**

<table>
<tr><th>Item</th><th>Evidence gathering activities, evidence checked, and assessment conclusion</th></tr>
<tr><td></td><td>
<table><tr><td>Kelwa Buzurg Village, Punasa Tehsil, Khandawa district of Madhya Pradesh</td><td>05-December-2023</td></tr></table>
Assessment team verified that; project activity was set-up in Madhya Pradesh state in India local stakeholder meeting has been conducted at time and date, same has been provided above, assessment team has checked the stakeholder meeting, invitation letter, MoM, meeting photos, attendance sheet, and other evidence and conformed that all the evidence is found consistent with interview outcomes. Assessment team verified that; during document review they have checked the evidence and it is conformed that, invitation sent through public notice and telephone communications on dates 20-November-2023 – local community members were invited through public notice while local government, local NGO workers and media personnel were invited mostly by telephonic communications. Project activity is located in several location of host country India and the Language used in invitation to stakeholders included both English and Hindi.
Thus, accepted by assessment team.
</td></tr>
<tr><td>Consultation outcome</td><td>
The meeting location – Kelwa Buzurg Village site Location
Assessment team found that, there were some suggestions received from respondents, Project proponent has incorporated the same in while preparing the documents.
The purpose of the meeting was clearly outlined i.e., making local stakeholders including employees of the company, local administration representatives, local community, NGOs working for environmental cause etc., gets the awareness of this solar project activity that involves electricity generation using solar energy through solar PV technology. The primary objective of this project is generating the clean energy by reducing emissions of GHGs in comparison to pre-project or baseline scenario.
Assessment team verified that; During the local stakeholder consultation, suggestions were gathered and analyzed. Same has been provided to the assessment team, the Project Participant (PP) documented these inputs and implemented appropriate actions, which were reviewed for alignment with stakeholder feedback. The assessment team verified this by cross-checking stakeholder meeting
</td></tr>
</table>

Item	Evidence gathering activities, evidence checked, and assessment conclusion
	minutes, action plans, and on-site observations to ensure transparency and adherence to agreed measures.
Ongoing communication	Assessment team has checked the DPR and LSC report and it is found that, stakeholder communication is continuous process, PP has maintained the grievance register at the project site or any employee or stakeholder can lodge a complaint or query through the register or via email. Same has been verified during the site visit and interview with site in-charge and stakeholder as well. Thus, find accepted.
Stakeholder input	The solar project listened to suggestions, concerns, and opinions of stakeholders and improved the results of the environmental impact study, and took comprehensive measures to prevent environmental impacts and benefit the community. Same was checked with the CoP report. CL 01 was raised on this section and closed successfully. Please refer Appendix 3 for further details.

3.2.1.3 Free, Prior, and Informed Consent

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Obtaining consent	The solar power plant is floating solar plant installed at Omkareshwar water reservoir with no utilisation of any indigenous land or resources. Project proponent has obtained all the necessary approvals from the government before commissioning of the project. Further, no dispute/concerns were raised by the stakeholders during stakeholder meetings. Assessment team checked the same with detailed project report and also verified it through interviews with the local people during Validation site visit.
Outcome of FPIC discussion	Assessment team found that, there were some suggestions received from respondents. Project proponent has incorporated the same VCS PD and project proponent has bound to fulfilled the report. Assessment team verified it through interviews with the local people during Validation site visit.

3.2.1.4 Grievance Redress Procedure

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Development process	Assessment team verified that; Grievance/Feedback register is available at project site office where any stakeholder can come and register his/her feedback along with their particulars. That will be handle by site in-charge and further reviewed by management team. Assessment team checked the same as mentioned in detailed project report also verified it through interviews with the project proponent during Validation site visit.
Grievance redress procedure	Grievance redressal mechanism is established as per company policy and assessment team verified that; any complaint register has been timely resolved and mean time it is informed to the concerned complainer who has raised the query. Assessment team checked the same and found appropriate.

3.2.1.5 Public Comments

Comments received	Actions taken by the project proponent	Evidence gathering activities, evidence checked, and assessment conclusion
No comments received during the commenting period, as evident from the VCS pipeline in the web interface.	In accordance with the requirement in clause 3.18.9 of the VCS Standards version 4.7 “All VCS project is subject to a 30-day public comment period. The date on which project is listed on the project pipeline marks the beginning of the project’s 30 days public comment period. The PP listed their project activity in the VCS pipeline for 30 days from 13-Feburary-2024 to 14-March-2024[9] for public comment.	VVB team has checked the web interface and observed no comment received, during the 30 days.

[9] Verra Search Page

3.2.2 Respect for Human Rights and Equity

3.2.2.1 Labor and Work

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Discrimination and sexual harassment	Workers including women will be employed during the project period. All workers, both women and men, will be treated equally, with no discrimination or sexual harassment tolerated. Assessment team verified that; as per national law related to "The Sexual Harassment of Women at Workplace (Prevention, Prohibition and Redressal) Act, 2013[10]" Same was found in-line with the companies polices and interviews with the project proponent.
Management experience	The assessment team confirmed that the project management team possesses the necessary experience and expertise to effectively implement similar project activities. Evidence gathered through interviews, and project documentation indicates that team members have extensive experience in solar power installation, environmental protection, project planning, implementation, and supervision. The project manager has demonstrated expertise in engaging communities, supported by evidence of prior community outreach initiatives. The team's skills and experience were verified through background checks, professional references, and relevant certifications. The Project Proponent (PP) has implemented a comprehensive recruitment strategy to focusing on attracting professionals with expertise in solar power, environmental protection, and project management to enhance team capabilities. Additionally, the PP collaborates with partner organizations to leverage technical expertise and support for successful project execution. Based on the reviewed evidence, the management team is well-prepared to execute the project effectively, supported by a robust plan to further strengthen expertise as needed.
Gender equity in labor and work	The project proponent was required by the parent company and local government to ensure gender equality in labor and work in the same line with parent company's discrimination and sexual harassment related policy. Assessment team has checked the company human right policy and found it consistent with ministry of labour and employment[11].

[10] https://doe.gov.in/files/inline-documents/DoE_Prevention_sexual_harassment.pdf

[11] Labour Law Reforms | Ministry of Labour & Employment|Government of India

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Human trafficking, forced labor, and child labor	Assessment team verified that; project proponent does not and will not use victims of human trafficking, forced labor, and child labor. The solar project operates in compliance with the ministry of labour and employment[12] and does not employ victims of human trafficking, forced labor, or child labor, as required by the Act. CL 01 was raised on this section and closed successfully. Please refer Appendix 3 for further details.

3.2.2.2 Human Rights

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Human rights	The assessment team verified that there have been no violations of any member's human rights and that the project proponent has complied with the human rights of all employees and local stakeholders. The project is a solar power plant. The assessment team additionally affirms that, in compliance with the protection of the human right act 1993 of India[13] , the project proponent consulted with local stakeholders in order to protect and advance the rights of intellectual property, customary right holders, and LCs. Same is found consistent with code of practice of project activity.

3.2.2.3 Indigenous Peoples and Cultural Heritage

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Preservation and protection of cultural heritage	The assessment team verified that project installed at the Omkareshwar water reservoir and no indigenous people lived close to the floating solar power plant, and that no cultural heritage site existed in the vicinity of the project activity. These results were further verified during the site visit and interviews with the project proponent and local stakeholders.

[12] Labour Law Reforms | Ministry of Labour & Employment|Government of India

[13] The Protection of Human Rights Act, 1993 | National Human Rights Commission India (nhrc.nic.in)

3.2.2.4 Property Rights

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Rights to territories and resources	Not applicable as this is a floating solar project and 100% owned by the project legal owner. Hence, there are no any legal or customary tenure/access rights to territories, property, and resources, including collective and/or conflicting rights, held by stakeholders. Same has been verified during document review and site visit and it is found consistent thus accepted.
Respect for property rights	Not applicable as this is a floating solar project and 100% owned by the project legal owner. Hence, there are no any legal or customary tenure/access rights to territories, property, and resources, including collective and/or conflicting rights, held by stakeholders. Same has been verified during document review and site visit and it is found consistent thus accepted.

3.2.2.5 Benefit Sharing

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Process used to design the benefit sharing plan	The project activity involves utilizing solar energy to generate power for internal usage. Considering that the project proponent is using this solar facility for electricity generation and supplied to grid. Since it is not a community-based initiative, there are no requirement of benefits sharing from it. However, project proponent during the local stakeholder consultation process made sure that the local community, other stakeholders, and probable project impacted individuals (PAPs) were notified beforehand and given the opportunity to actively engage and be consulted. Due to the open nature of the consultation process, this helps to lessen the sense of uncertainty among the local population and possible PAPs.
Summary of the benefit sharing plan	Assessment team verified that; The Omkareshwar Dam Reservoir's proposed floating solar photovoltaic plant in the Kaveri branch is probably going to benefit the local social structure in terms of the economics and community development initiatives. In the short and long term, the project will expand local job prospects based on education and skill levels, raise tax income,

	and create demand for goods and services through local contracting. However, this does not applicable to the benefit sharing.
Approval and dissemination of benefit sharing plan	Under the Government of India's Solar Park Scheme, the Floating Solar Park will be constructed and eligible for all of the financial advantages offered by the program. Under the "Scheme for Development of Solar Parks and Ultra Mega Solar Power Projects," MNRE will award a grant to support the construction of the parks.

3.2.3 Risks to Local Stakeholders and the Environment

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Risks to stakeholder participation	The assessment team verified that the project activity comprised floating solar electricity generation and that no concerns had been identified. The stakeholder consultation documents and the local stakeholder gathering also makes this prominently apparent. The VVB team states that by guaranteeing a more extensive and comprehensive stakeholder interaction channel, the project proponent has reduced this risk. To address any complaint or questions, PP developed the grievance redressal mechanism. Additionally, the project proponent is bound with company policy and government regulation to comply with to the Practices. Same was confirmed during site visit interviews with PP & local stakeholders.
Working conditions	Employee productivity and well-being are affected by a wide range of elements that are included in the environment of work. These elements include workplace organization, surroundings, and physical elements. Through the project activity's rules and regulations, the VVB team verified that the working conditions were determined to be in compliance with the labor laws of the country. Same was confirmed during site visit interviews with employees.
Safety of women and girls	Assessment team verified that, Within the project, the project activity provides women with opportunities for employment in management and operational roles. The project activity will keep track of the number of female staff. Moreover, assessment team confirm that, the committee has been entrusted by the project proponent to investigate any complaints made by female employees on any matter. Same has been find consistent with company policy. The project owner guarantees equal pay rights, prohibits any negative effects, and abides by a non-discrimination policy with regard to hiring and salary. This was verified by speaking with the project activity's

Item	Evidence gathering activities, evidence checked, and assessment conclusion
	monitoring staff during an on-site visit. Further, Same was found consistent with the company's policy.
Safety of minority and marginalized groups, including children	Assessment team verified that, project activity has followed the community engagement, cultural sensitivity, equal access & opportunity, anti-discrimination policy, child protection policy, same was observed at the time of site visit of the project activity. Further, no child labour was observed at the site.
Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials)	The project activity is solar project plant, there is no air, noise, or hazardous waste generated at the project site, same was observed at the time of site visit, VVB team verified that, project activity compliance with the law of the host country[14]. CL 01 was raised on this section and closed successfully. Please refer Appendix 3 for further details.

3.2.4 Ecosystem Health

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Impacts on biodiversity and ecosystems	Assessment team reviewed the Final Environmental and Social Impact Assessment (ESIA) Report prepared by Rewa Ultra Mega Solar Ltd. in May 2022 and verify that, the reservoir ecology has less amount of biodiversity and richness of species and the water system is not a natural flowing system. The organic content of the reservoir sediments also found low as the system can be considered as new and developed artificially after development of the Dam. The project is rooftop project it does not any impact on biodiversity and ecosystem. As the area does not fall under any flyway for migratory birds, the assemblage of aquatic and wetland birds was truncated. The richness of such birds in terms of abundance and diversity was low and patchily distributed. Thus, assessment team verified that, there is no impact on biodiversity and ecosystem due to the project activity. Same was confirmed at the time of the site visit and interview with the PP & local stakeholder.

[14] CPCB | Central Pollution Control Board

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Soil degradation and soil erosion	The assessment team confirmed that since, floating solar plants are positioned on bodies of water rather than on land, there is no soil erosion caused by them. Mounted on floating platforms that sit atop bodies of water, including lakes, ponds, or reservoirs, are floating solar panels. Soil erosion is effectively removed as a concern because there is no ground installation or soil disturbance involved. Furthermore, floating solar plants have a cooling effect that lowers water evaporation, increasing panel efficiency and conserving water. Same has been find During the site inspection. Thus, this is not affected.
Water consumption and stress	Assessment team verified that; there were not wasting of water for cleaning the solar panel, a moderate amount of water is needed to clean the solar panel. which is done about three times every month or more often depending on the weather. To ensure a sufficient supply of water, commercial organizations, and municipal authorities. Therefore, it is expected that there would be no impact from water consumption on the neighbouring villages. Thus, accepted by assessment team.
Usage of fertilizers	Assessment team verified that; project activity is solar power plant. There is no use of fertilizer in the solar project. No findings have been raised in this section.

3.2.4.1 Rare, Threatened, and Endangered species

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Species and habitat	The VVB team verified that no detrimental effects had been noticed in either IUCN-listed or non-listed species. The VVB team also confirmed that the project activity complies with all applicable environmental laws and permits requirements during the project activity's onsite inspection. Local stakeholders were interviewed, and the VVB team verified that neither the project's construction nor its execution will have an impact on their living situation, nor will the project's activities impact their day-to-day activities. According to the PP and the local representative and same has been conformed with the Ministry of environment forest and climate change (Botanical Survey of India). Thus, accepted by assessment team.

3.2.4.2 Introduction of Species

Species introduced	Evidence gathering activities, evidence checked, and assessment conclusion
Not applicable	Assessment team verified that; the floating solar project is located in Omkareshwar water reservoir in Punasa tehsil of Madhya Pradesh state of India and the site doesn't have any threatened plant and animal, or Marine species as confirmed from the MoEF&CC report[15]. Assessment team conformed the same with Environmental and Social Impact Assessment (ESIA) report for floating solar power plant prepared by third party Voyants Solutions Pvt. Ltd./20/ May 2022 and it is found that plant is not affecting any marine or terrestrial species.

Existing invasive species	Evidence gathering activities, evidence checked, and assessment conclusion
Not Applicable	VVB team verified that, during onsite visit VVB team found that there is no invasive species existed at the project location, same was conformed through the documents review i.e ESIA report/20/ and interview with local stakeholder as well. Thus, no findings have been raised in this section.

3.2.4.3 Ecosystem conversion

Item	Evidence gathering activities and evidence checked
Ecosystem conversion	The VVB team conducted a site visit and communicated with a local stakeholder who was present throughout the interview to confirm that there were no endangered species of plants or animals at the project site, nor were there any areas that would disrupt valuable ecosystems. Moreover, It is less invasive than traditional land-based solar farms, floating solar plant had developed with consideration for local aquatic ecosystems. Assessment team after reviewing the ESIA report/20/ it is clear that, solar plant ensures minimal disruption to aquatic life, plant diversity, and water quality.

3.3 Application of Methodology

3.3.1 Title and Reference

Methodology: ACM0002: - Version 22.0

Project Type: Type-I

Sectoral scopes: 1

Title: Grid-connected electricity generation from renewable sources

[15] https://moef.gov.in/moef/wildlife-wl/wildlife/index.html

The methodology refers to following CDM tools:

Tool 01: Tool for the demonstration and assessment of additionality- Version 07.0.0[16]

Tool 05: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation- Version 3.0[17]

Tool 07: Tool to calculate the emission factor for an electricity system - Version 07.0[18]

Tool 24: Common Practice – Version 03.1[19]

Tool 27: "Investment Analysis" Version 14.0[20]

CL 01 & CAR 02 were raised on this section and closed successfully. Please refer Appendix 3 for further details.

3.3.2 Applicability

Methodology: ACM0002: Grid-connected electricity generation from renewable sources - Version 22.0.

Methodology ID	Applicability condition	Assessment and conclusion
ACM0002: Grid-connected electricity generation from renewable sources - Version 22.0.	5. This methodology is applicable to grid-connected renewable energy power generation project activities that: a. Install a Greenfield power plant; b. Involve a capacity addition to (an) existing plant(s);	The solar plant installation of a greenfield floating solar power. The project activity will displace electricity from Indian Grid, which is supplied by predominantly fossil fuel plants. Solar PV projects generating electricity and supplied to grid.

[16] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-01-v7.0.0.pdf

[17] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-05-v3.0.pdf

[18] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf

[19] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-24-v1.pdf

[20] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v14.0.pdf

The assessment team confirmed that the project proponent had used the investment analysis version 12 for the investment analysis calculation, however tool 27 had been upgraded to version 13. Moreover, the existing investment analysis's version 12' grace period remains valid until July 25, 2024. Thus, this is accepted.

Methodology ID	Applicability condition	Assessment and conclusion
	c. Involve a retrofit of (an) existing operating plant(s)/unit(s); d. Involve a rehabilitation of (an) existing plant(s)/unit(s); or e. Involve a replacement of (an) existing plant(s)/unit(s); or f. Install a Greenfield power plant together with a grid-connected Greenfield pumped storage power plant. The greenfield power plant may be directly connected to the PSP or connected to the PSP through the grid.	
	7. In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Integrate BESS with a Greenfield power plant; (b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic1 or wind power plant(s)/unit(s); (c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any	Assessment team verified that; project activity supplied generated electricity to the grid and project activity does not involve the integration of BESS hence this condition is not applicable. VVB team confirm that, PP have explained scenarios appropriately in the VCS PD and found correct.

Methodology ID	Applicability condition	Assessment and conclusion
	other changes to the existing plant(s); (d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s); (e) Integrate a BESS together with a Greenfield power plant that is operating in coordination with a PSP. The BESS is located at site of the greenfield renewable power plant.	
	8. The methodology is applicable under the following conditions: (a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit must have started commercial operation prior to the start of a minimum historical reference period of five years. The reference period is used for the	VVB team after conducting onsite visit and document review it is conformed that; The project activity involves construction and operation of greenfield grid-connected floating solar power project. Hence, complies to this applicability condition (a). Since the project activity does not include capacity additions, retrofits, rehabilitations, or replacements of existing plant/unit the applicability condition (b) is not applicable/ relevant for the project activity. The project activity does not involve the integration of BESS hence this condition (c) is not applicable. The project activity does not involve the integration of BESS hence this condition (d) is not applicable.

Methodology ID	Applicability condition	Assessment and conclusion
	calculation of baseline emissions and defined in the baseline emission section. Furthermore, no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity; (c) In case of Greenfield project activities applicable under paragraph 7(a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g., by referring to feasibility studies or investment decision documents); (d) The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies 2 may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil	The project activity does not involve the PSP, hence this condition (e) is not applicable.

Methodology ID	Applicability condition	Assessment and conclusion
	fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g., week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period. (e) In case the project activity involves PSP, the PSP shall utilize the electricity generated from the renewable energy power plant(s) that is operating in coordination with the PSP during pumping mode	
	9. In case of hydro power plants, one of the following conditions shall apply. (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and	The VVB team verified that this applicability condition is not applicable or relevant to the project activity because it relates to hydro power projects. The project activity entails building and running a greenfield grid-connected floating solar power project that uses solar energy to generate electricity.

Methodology ID	Applicability condition	Assessment and conclusion
	the power density, calculated using equation (7), is greater than 4 W/m2 ; or (c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m2 ; or (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m2 , and all of the following conditions shall apply: i. The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m2 ; ii. Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; iii. Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m2 are: a. Lower than or equal to 15 MW; and	

Methodology ID	Applicability condition	Assessment and conclusion
	b. Less than 10 per cent of the total installed capacity of integrated hydro power project.	
	10. In the case of integrated hydro power projects, project participants shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or (b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and	Assessment team verified that; Since this applicability condition relates to hydro power projects, it is not applicable or relevant to the project activity, which entails building and running a greenfield grid-connected floating solar power project that uses solar energy to generate electricity.

Methodology ID	Applicability condition	Assessment and conclusion
	rainfall for minimum of five years prior to the implementation of the CDM project activity.	
	11. In the case of PSP, the project participants shall demonstrate in the PDD that the project is not using water which would have been used to generate electricity in the baseline.	This isn't a PSP project. This criterion is thus not relevant.
	12. The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units.	VVB team verified that; Since the project's activity entails building and running a greenfield, grid-connected floating solar power project that uses solar energy to generate electricity, this applicability requirement is irrelevant, just as it is for moving away from fossil fuels and toward renewable energy sources or biomass-fired power plants.
	13. In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is "the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project	VVB team verified that; Since the project activity entails building and running a greenfield grid-connected solar power plant that uses solar energy to generate electricity, the application criterion "10" is irrelevant because it also applies to capacity expansions, retrofits, rehabilitations, and replacements.

Methodology ID	Applicability condition	Assessment and conclusion
	activity and undertaking business as usual maintenance".	
	14. In addition, the applicability conditions included in the tools referred to below apply.	VVB team verified that; The applicability of the tools is outlined below

TOOL 1: - Tool for the demonstration and assessment of additionality - Version 7.0.0 (EB 70, Annex 08)

Tool ID	Applicability Criterion	Assessment and conclusion
TOOL01: Version 07.0.0	Para 09: The use of the "Tool for the demonstration and assessment of additionality" is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	VVB team verified that; This condition is not applicable as the project participant is not proposing new methodologies.
TOOL01: Version 07.0.0	Para 10: Once the additionally tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	VVB team verified that; Since this additionality tool is being used in project activities, this criterion is relevant.

TOOL 5: - Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation version 3.0.

Tool ID	Applicability Criterion	Assessment and conclusion
TOOL 5 Version 3.0	If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption: *(a)* Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer; *(b)* Scenario B: Electricity consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or	VVB team verified that, the project activity used the electricity from the grid, before commissioning the power plant and the proposed solar plant it is connected to grid. Thus, project proponent has used the scenario A.

Tool ID	Applicability Criterion	Assessment and conclusion
	(c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid.	
TOOL 5 Version 3.0	This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated: (a) Scenario I: Electricity is supplied to the grid; (b) Scenario II: Electricity is supplied to	VVB team verified that, the project activity is grid connected or supplying electricity to client through the grid thus, scenario I & II is not applicable, hence it is captive consumption solar power plant thus scenario III is applicable in this section.

Tool ID	Applicability Criterion	Assessment and conclusion
	consumers/electricity consuming facilities; Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities.	
TOOL 5 Version 3.0	This tool is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. The tool only accounts for CO_2 emissions.	VVB team verified that, as per baseline scenario status, captive renewable power production technologies do not exist.

TOOL07: Tool to calculate the emission factor for an electricity system version 7.0

Tool ID	Applicability Criterion	Assessment and conclusion
Tool 07 Version 7.0	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid	VVB team verified that, this section is related to grid connection of the project activity, However the project activity used fossil fuel dominated grid power that's being used prior to the project activity, this scenario is applicable in present scenario.
Tool 07 Version 7.0	Under this tool, the emission factor for the project electricity system can be	VVB team verified that, project activity used electricity from the grid which is mainly

Tool ID	Applicability Criterion	Assessment and conclusion
	calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project Proponent, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in "Appendix 1: Procedures related to off-grid power generation" should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity	dominated by fossil fuel as baseline scenario, as per applied methodology calculation of the emission factor has been provided in section in 4.1 of the project description and same is reflecting in 3.3.4 of the validation report. Thus, this toll is applicable.
Tool 07 Version 7.0	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	Not applicable: - VVB team verified that, this section is not applicable thus the project activity is located in India since India is non-annex I country.

Tool ID	Applicability Criterion	Assessment and conclusion
Tool 07 Version 7.0	Under this tool, the value applied to the CO_2 emission factor of biofuels is zero.	VVB team verified that, solar power project is not involve any kind of biofuels, thus this scenario is not use-full.

TOOL24: Common Practice – Version 03.1 (EB 84, Annex 07)

Tool ID	Applicability Criterion	Assessment and conclusion
TOOL24: Version 03.1	Para 03: This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, or baseline and monitoring methodologies that use the common practice test for the demonstration of additionality.	Assessment team verified that; Methodological tools such as "Tool for the demonstration and assessment of additionality," "Combined tool to identify the baseline scenario and demonstrate additionality," or baseline and monitoring methodologies that employ the common practice test for the demonstration of additionality are being applied in the proposed project activity.
TOOL24: Version 03.1	Para 04: In case the applied approved baseline and monitoring methodology defines approaches for the conduction of the common practice test that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	Assessment team verified that; this methodological tool describes techniques for conducting the common practice test that are not defined by the applied authorized baseline and monitoring methodology.

TOOL27: Investment Analysis version 14.0

Tool ID	Applicability Criterion	Assessment and conclusion
Tool 27 Version 14.0	This methodological tool is applicable to project activities that apply the methodological tool "Tool for the demonstration and assessment of additionality", the methodological tool "Combined tool to identify the baseline scenario and demonstrate additionality", the guidelines "Non-binding best practice examples to demonstrate additionality for SSC project activities", or baseline and monitoring methodologies that use the investment analysis for the demonstration of additionality and/or the identification of the baseline scenario.	VVB has cross checked the project description and IRR Sheet and confirms that the applicability criterion is met as the project uses the methodological tool 27 "Investment analysis.
Tool 27 Version 13.0	In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	This condition is not applicable. The applied methodology, ACM0002, version 22.0 does not contain any requirements for the investment analysis that are different from those described in this methodological tool.

CL 01 & CAR 02 were raised on this section and closed successfully. Please refer Appendix 3 for further details.

3.3.3 Project Boundary

As per ACM0002, Version 22.0 "The spatial extent of the project boundary includes the project power plant / unit and all power plants / units connected physically to the electricity system that the project power plant is connected to. The project boundary therefore includes the project site where the power plant has been installed, associated power evacuation infrastructure, energy metering points, switch yards the national grid of India. The boundary also extends to the project power plant and all power plants connected physically to the electricity system that the VCS project power plant is connected to."

To define the project boundary, the solar project identifies all potential greenhouse gas emission sources, sinks, and reservoirs within the project boundary. This includes the electricity generation from the grid, electricity generation as part of the baseline, and the solar project area for project activity.

VVB team has reviewed the following document during site visit and document review the details of documents are as follows: -

- Commissioning certificate
- Detailed Project Report
- Ownership Records
- Employment records
- Third Party PLF Report
- Meter calibration Report
- Plant as built layout

Each component identified during the site visit is found to be in line with the submitted evidence.

The spatial extent of the project boundary includes industrial, commercial facilities consuming energy generated by the system. In the case of electricity generated and supplied to distributed users (e.g. residential users) via mini/isolated grid(s) the project boundary may be confined to physical, geographical site of renewable generating units. The boundary also extends to the project power plant and all power plants connected physically to the electricity system to which the project power plant is connected.

The project activity will displace electricity from Indian Grid, which is supplied by predominantly fossil fuel plants. The project activity instances would replace equivalent quantum of fossil fuel dominated grid power that's being used prior to the project activity instances.

Source		Gas	Included?	Assessment and conclusion
Baseline	Natural Gas based captive electricity generation	CO_2	Yes	VVB team confirm that, the project has included CO_2 source as main emission source and is conservative & in line with methodology. This amount of emission reduction will be monitored as per monitoring plan in the Project description.
		CH_4	No	VVB team verified that, methane emission i.e. CH_4 is not included in project activity being conservative & in line with methodology. Thus, not applicable.

Source		Gas	Included?	Assessment and conclusion
		N_2O	No	VVB team verified that, Nitrous oxide emission i.e. N_2o is not included in project activity conservative & in line with methodology. Thus, not applicable
		Other	No	Not applicable
Project	CO_2 emissions from the consumption of grid electricity used as auxiliary power to the project activity	CO_2	Yes	The VVB team verified that CO_2 emissions from the consumption of grid electricity used as auxiliary power have been appropriately identified as the main emission source for this Greenfield Solar PV Power Project activity. These emissions will be monitored and accounted for in accordance with the monitoring plan provided in the Project Description (PD).
		CH_4	No	VVB team verified that, methane emission i.e. CH4 is not included in project activity. Thus, not applicable.
		N_2O	No	VVB team verified that, Nitrous oxide emission i.e. N2O is not included in project activity as no project emissions associated with project activity & in line with meth. Thus, not applicable

VVB team verified that, details of the project boundary and each selected sources, sinks and reservoirs are appropriately justified for the project activity.

3.3.4 Baseline Scenario

As per the approved consolidated methodology ACM0002, Version 22.0, para 27, If the project activity is the installation of a Greenfield power plant with or without a BESS, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the "TOOL07: Tool to calculate the emission factor for an electricity system".

The project activity involves setting up of solar projects to harness the solar energy to produce electricity and supply to the grid. In the absence of the project activity, the equivalent amount of power would have been supplied by the Indian grid, which is fed mainly by fossil fuel fired plants. Hence, the baseline for the project activity is the equivalent amount of power from the Indian Grid.

The combined margin ($EF_{grid,CM,y}$) is the result of a weighted average of two emission factor pertaining to the electricity system: the operating margin (OM) and build margin (BM), in accordance with the TOOL7: Tool to calculate the emission factor for an electricity system – Version 07 Calculations for this combined margin must be based on data from an official source (where available) and made publicly available. In India, Central Electricity Authority (CEA), Government of India provides this data, and accordingly the same has been used.

Calculations for this combined margin must be based on data from an official source (where available) and made publicly available. The CEA database version 19 is the latest available data & the same is considered for emission factor calculations.

The baseline emissions are calculated as follows:

$BE_y = EG_{BL,y} \times EF_{CO2,y}$

Where:

BE_y = Baseline emissions in year y (tCO_2/yr)

$EG_{BL,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EF_{CO2,y}$ = Emission factor(tCO_2/MWh)

Grid Emission Factor Calculation: -

The combined margin of the INDIAN National Grid used for the project activity is as follows:

Parameter	Value	Nomenclature	Source
$EF_{grid,CM,y}$	0.9352 tCO_2/MWh	Combined margin CO2 emission factor for the project electricity system in year y	Build margin CO_2 emission factor for the project electricity system in year y Baseline CO_2 Emission Database[21], Version 19.0. November 2023 published by Central Electricity Authority (CEA), Government of India.
$EF_{grid,OM,y}$	0.9580 tCO_2/MWh	Operating margin CO2 emission factor for the project electricity system in year y	Build margin CO_2 emission factor for the project electricity system in year y Baseline CO2 Emission Database[22], Version 19.0. November 2023 published by Central Electricity

[21] https://cea.nic.in/cdm-co2-baseline-database/?lang=en

[22] https://cea.nic.in/cdm-co2-baseline-database/?lang=en

			Authority (CEA), Government of India.
$EF_{grid,BM,y}$	0.8670 tCO_2/MWh	Build margin CO2 emission factor for the project electricity system in year y	Build margin CO_2 emission factor for the project electricity system in year y Baseline CO_2 Emission Database[23], Version 19.0. November 2023 published by Central Electricity Authority (CEA), Government of India.

Assessment team conclude that the above-mentioned baseline scenario is in line with the applied methodology "ACM0002: Grid-connected electricity generation from renewable sources - Version 22.0". Same is also reflected in final PD version 05. Thus accepted.

The combined margin emissions factor is calculated as follows:

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times w_{OM} + EF_{grid,BM,y} \times w_{BM}$$

Equation (16)

Where:

$EF_{grid,BM,y}$ = Build margin CO_2 emission factor in year y (t CO_2/MWh)

$EF_{grid,OM,y}$ = Operating margin CO_2 emission factor in year y (t CO_2/MWh)

w_{OM} = Weighting of operating margin emissions factor (per cent)

w_{BM} = Weighting of build margin emissions factor (per cent)

$EF_{grid,CM,y}$ = 0.9580*0.25 + 0.8670*0.75

= 0.9352 (tCO_2/MWh)

3.3.5 Additionality

For demonstrating additionality under VCS, the project activity is required to undergo the following tests

a) **Legal Requirement Test:** - based on the following available literature on Electricity Market Law in India: -

[23] https://cea.nic.in/cdm-co2-baseline-database/?lang=en

- Electricity Act 2003 (May 2007 Amendment)[24]
- National Electricity policy 2005[25]
- The Electricity (Supply) Act, 1948[26]
- The Electricity Regulation Commission Act, 1998[27]
- Schedule 1 of Ministry of Environmental and Forest notification[28]
- National Renewable Energy Act 2015[29]

  The Project activity conforms to all the applicable laws and regulations in India

- Power generation using renewable energy is not a legal requirement or a mandatory option.
- The Indian Electricity Act, 2003[/40/] (May 2007 Amendment) does not influence the choice of fuel used for power generation.
- There is no legal requirement on the choice of a particular technology for power generation.
- Both the baseline and project activity are in compliance with laws and regulations required. There is no mandatory requirement to implement the project activity.

b) **Additionality Tests:**

As per the applied CDM methodology ACM0002 version 22.0[/08/], additionality of the following project activity is demonstrated and assessed by the latest version of Tool 01: "Tool for the demonstration and assessment of additionality" Version 07.0.0[/08/]. The PP has adopted the stepwise approach for demonstrating and assessing the additionality of the project activity as follows:

The project was envisaged for total capacity of 100 MW_{ac} in Madhya Pradesh state of India. Currently, Project activity is fully commissioned and continuously contributing towards emission reduction. VCS PD[/04/] for this project activity was web-hosted for public consultation 13-Feburary-2024 to 14-March-2024. Start date of the Project is 06-June-2024 which is Commissioning[/01/] date of 100ac MW plant site.

[24] https://cercind.gov.in/Act-with-amendment.pdf

[25] https://powermin.gov.in/en/content/national-electricity-policy

[26] https://cercind.gov.in/ElectSupplyAct1948.pdf

[27] https://cercind.gov.in/ElectReguCommiAct1998.pdf

[28] https://parivesh.nic.in/writereaddata/MINISTRY%20OF%20ENVIRONMENT%20AND%20FORESTS%20SO474E.pdf

[29] https://policy.asiapacificenergy.org/sites/default/files/Draft%20National%20Renewable%20Energy%20Act%202015.pdf

The demonstration of additionality for the proposed Project activity is being carried out in accordance with the additionality tool provided by the UNFCCC i.e., Tool 01 "Tool for demonstration and assessment of Additionality" Version 7.0/08/. The tool provides a step-wise approach to establish additionality of the project activity has been followed, details of which are provided in the following paragraphs: The tool provides a step-wise approach to demonstrate and assess the additionality of a project (figure below). These steps are:

(a) STEP 0. Demonstration that a proposed project activity is the first-of-its-kind.

(b) STEP 1. Identification of alternative scenarios.

(c) STEP 2. Barrier analysis.

(d) STEP 3. Investment analysis (if applicable).

(e) STEP 4. Common practice analysis.

As per of TOOL 01: 'Tool for demonstration and assessment of Additionality' version 7.0:/08/ -

Step 0: Demonstration whether the proposed project activity is the first of-its-kind.

The project activity is a large-scale solar power project undertaken in Indian state of Madhya Pradesh. This is not the first such project to be installed in the country or in the state and therefore project activity does not meet this criterion.

Step 1: Identification of alternatives to the project activity consistent with current laws and regulations

As per the applied CDM methodology/08/ paragraph 27, the project activity is the installation of a Greenfield power plant, and the baseline scenario is that the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid. Thus, the baseline scenario is applied as per the CDM methodology.

Sub-step 1a: Define alternatives to the project activity: Identify realistic and credible alternative(s) available to the project participants or similar project developers that provide outputs or services comparable with the proposed project activity.

The purpose of the project activity is aimed to generate power from the solar power plant using renewable energy (solar) resources and feed the electricity generated to the grid. Hence, the following alternatives are considered:

Alternative 1: The proposed project activity undertaken without being registered as a Proposed project activity.

The option of proposed project activity being undertaken without being registered as a project activity is in compliance with all applicable legal and regulatory requirements and can be a part of the baseline. However, this is not a plausible option as deployment of the project activity faces significant financial barriers established in the subsequent section. The electricity generated from the project activity would have been fed to the India's National Grid, which has a number of power plants including existing and upcoming ones which would be used in the absence of project activity. In the absence of project activity, power from the same grid will continue to provide for the requirement of the country.

Alternative 2: Continuation of the current situation (No proposed project activity and equivalent amount of energy would have been produced by the grid electricity system through its currently running power plants and by new capacity addition to the grid).

In such case the project owner would have continued without investment in project activity and continued with its usual business activities. In such case grid would continue with the fossil fuel-based power projects and this would result in GHG emissions. Hence, existing power plant and the new capacity add-on from a fossil fuel-based power plant is appropriate, realistic & credible baseline alternative for the project activity.

Outcome of Sub-step 1a: Identified realistic and credible alternative scenario (s) to the project activity.

Of the two alternatives outlined above, the first alternative is not possible as project activity is not viable without carbon credit benefits and second alternative is the baseline scenario for the project activity as per methodology. Therefore, continuation of the current situation is the likely alternative to the project activity.

The project being a green field project activity the baseline scenario in line with the methodology is "the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the "Tool to calculate the emission factor for an electricity system Version 7.0"

Sub-step 1b: Consistency with mandatory laws and regulations: The alternative(s) shall be in compliance with all applicable legal and regulatory requirements, even if these laws and regulations have objectives other than GHG reductions, e.g. to mitigate local air pollution.

Installation of a large-scale solar power project including the one proposed as project activity is consistent with mandatory laws and regulations at national (India) and sub-national level (state of Madhya Pradesh). The environmental regulations, legislations and policy guidelines in respect to the project activity is governed by Ministry of Environment, Forest and Climate Change (MoEF&CC), Delhi supported by Central Pollution Control Board (CPCB). In accordance to Annexure-II MOEF&CC, OM on J-11013/41/2006-IA. II (I) dated on 07-july-2017 Solar Photovoltaic Power Projects are not covered under the ambit of EIA Notification, 2006 and its subsequent amendments and hence, it does not require preparation of EIA and pursuing Environmental Clearance from MoEF&CC.

Moreover, Solar Power Project is considered under “white category” for Consent to Establish/Operate as per regulation, and thus there is no necessity of obtaining the Consent to Operate’’. Since project activity falls under white category and the non-polluting in nature, the project fulfils the compliance to the local environmental laws and regulations.

Overall, the project activity is consistent with the following mandatory laws and regulations.

- Electricity Act, 2003 and its subsequent amendment as it does not influence the choice of fuel used for power generation nor has it mandated power generation using renewable energy only.
- Environmental (Protection) Act, 1986 and amendment(s)
- Environmental Impact Assessment (EIA) Notification, 2006 and amendment(s)
- The Air (Prevention and Control of Pollution) Act, 1981 including Rules 1982 and 1983 and amendment(s)
- The Water Prevention and Control of Pollution), Cess Act, 1977 including Rules 1978 and 1991 and amended in March, 2003
- Noise Pollution (Regulation and Control) Amendment Rules, 2017
- Hazardous and other Wastes (Management & Transboundary Movement) Rules, 2016 and amended in November, 2021
- E-waste (Management and Handling) Rules, 2020
- Bio-Medical Waste (Management and Handling) Rules 2016
- Plastics Waste Management Rules, 2016 amended in 2021
- Batteries (management and handling) rules 2019

Outcome of Sub-step 1b: Identified realistic and credible alternative scenario(s) to the project activity that are in compliance with mandatory legislation and regulations taking into account the enforcement in the region or country and EB decisions on national and/or sectoral policies and regulations.

Hence, both the alternatives enlisted above are found to comply with the mandatory laws and regulations taking into account the enforcement of the legislations in the region or country and EB decisions on national and/or sectoral policies and regulations. However, Alternative 2 “Continuation of the current situation” has been selected as the appropriate baseline alternative for this project activity in line with methodology.

Step 2: Investment analysis

In this section it is demonstrated that the project activity is not financially feasible without the revenue from the sale of VERs. This is demonstrated in following sections as per tool 27 "Investment analysis" (Version 14.0)/08/.

The project is bagged by AMP Energy Green Private Limited, through bidding/21/ via Reverse-bidding processes by Rewa Ultra Mega Solar Limited (RUMS) (RfP dated 02-November-2021)/21/ and reverse bidding date 06-May-2022. PP has considered the investment decision date as board resolution date 08-December-2021.

Following are the chronological events of the project activity.

S.l No.	Event	Date
1	RFP (Request for Proposal)[30] for Development of Floating Solar Park at Omkareshwar Reservoir, Madhya Pradesh by RUMSL (Rewa Ultra Mega Solar Limited)[31]	02-November-2021
2	Investment Note (formed by the board of M/s AMP Energy Green Seven Private Limited based on their market survey outcomes)	08-December-2021
3	Bid submission date	26-April-2022
4	Reverse auction date	06-May-2022
5	Release of Letter of Award	30-June-2022
6	Power Purchase Agreement between M.P. Power Management Company Limited and RUMSL and AMP Energy Green Seven Private Limited	04-August-2022
7	First Work Order released for string combiner boxes	03-November-2022
8	Date of Commissioning	06-June-2024

Sub-step 2a: Determine appropriate analysis method.

Project owner had demonstrated that the financial returns of the proposed VCS project activity would be insufficient to justify the required capital investment as per VCS Standard V.4.7.

[30] https://jmkresearch.com/wp-content/uploads/2021/11/RUMSL-600-MW-Floating-Solar-Madhya-Pradesh-Nov-2021.pdf

[31] http://rumsl.mp.gov.in/

The project is generating revenue in terms of power generated from the solar power plant being used for sell to national grids. Thus, simple cost analysis (Option I) is not appropriate. Hence out of 2 options left, investment comparison analysis (Option II) and benchmark analysis (Option III), benchmark analysis is used for the project activity as per project type and decision-making context. Therefore, the expected return on equity is considered appropriate benchmark.

All the steps followed to reach the conclusion have been assessed and the choice of analysis technique is accepted by the VVB team.

Sub-step 2b: Option III. Apply benchmark analysis:

Benchmark selection and its appropriateness:

As per Paragraph 15 of the tool 27 "investment analysis"[/08/], version 11.0 "The applied benchmark shall be appropriate to the type of IRR calculated. Local commercial lending rates or WACC are appropriate benchmarks for a project IRR. Required/expected returns on equity are appropriate benchmarks for an equity IRR. Benchmarks supplied by relevant national authorities are also appropriate. The DOE shall validate that the benchmarks used are applicable to the project activity and the type of IRR calculation presented".

The Project owner has chosen Post tax equity IRR as the financial indicator, based on the above the appropriate benchmark is required/expected returns on equity which is correctly chosen by the project owner, and it is acceptable.

As per paragraph 19 of the tool 27 Investment Analysis[/08/], version 11.0 "If the benchmark is based on parameters that are standard in the market, the cost of equity should be determined either by: (a) selecting the values provided in Appendix; or by (b) calculating the cost of equity using CAPM". Project owner has taken the default value for expected return on equity of 10.55% as given in the table 01 of Appendix of Tool 27- Investment Analysis (EB 116 Annex 2) Version 11.0[/08/] which was the latest applicable version at the time of submission of project activity for additionality demonstration. Hence the value considered by the project owner is appropriate and acceptable to verifier's team.

The benchmark return on equity in the tool is expressed in real terms. The post-tax equity IRR calculated is in nominal terms as escalation is considered in O&M cost. Accordingly, Project owner converted the default benchmark which is in real terms into nominal terms by using the following equation:

Nominal Benchmark[32] = {(1+Real Benchmark) *(1+Inflation rate)}-1.

VVB team referenced the book 'Corporate Finance" 2nd edition, by Aswath Damodaran[/22/]. In page 320 of the book, the same equation is mentioned for converting real into nominal values.

[32] As per Pg. 320 of Corporate Finance, Second Edition of Aswath Damodaran

Hence the assessment team considers the above equation as appropriate for converting real benchmark into nominal benchmark.

As per paragraph 16 of the Tool 27/08/ states that the inflation rate shall be obtained from the inflation forecast of the central bank of the host country for the duration of the crediting period, accordingly project owner has chosen the Reserve Bank of India (RBI) as Central Bank of host country (India) and it is India's monetary authority which is acceptable to the VVB team. The WPI inflation forecasted by RBI[33] for next 10 years minimum inflation rate published by the Reserve bank of India to be 4.00%.

Hence the nominal Benchmark estimated as = {(1+10.55%) *(1+4.00%) -1} = 14.97%.

The VVB team has verified the sources and confirmed that the benchmark identified to compare the financial attractiveness of the bundled project activity is appropriate.

Default Value for India as per UNFCCC guidelines	10.55%
Inflation targets as per RBI for 10 years	4.00%[34]
Nominal Benchmark	14.97%

b) Parameters and assumptions used:

The input parameters in the financial analysis/08/ have been taken as per the values and assumptions applicable and available at the time of decision to invest in the project activity in line with para 10 of tool 27 "investment analysis" version 11.0/08/. All the input values are sourced from the investment note/37/ prepared by the PP and were available at the time of decision making.

In 100 MWac floating solar project site, the investment note preparation date is 08-December-2021. The VVB team is convinced that the input parameters used in the detailed project report/10/ were valid and applicable at the time of investment decision.

The VVB team cross checked the input values with publicly available sources like CERC tariff order, Income Tax/Companies Act for its appropriateness at the time of the investment decision according to the requirement against VVS Paragraph 99. The assessment involved assessing the input data taken from Investment note, Purchase order, loan Sanction letter/23/, Income Tax Act, adoption of correct accounting principle and arithmetical accuracy. CARs and CLs were raised on non-conformities, and they were set right. With the corrections been incorporated, the input values considered appear to be in order. All input parameters examined in computation, on the basis of accuracy and appropriateness, are listed in the table below, along with VVB team assessment. The VVB team follows the guidelines outlined in para 99 and 101 of VVS/6/ version

33 Reserve Bank of India - Publications (rbi.org.in)

34 Reserve Bank of India - Publications (rbi.org.in)

3.0. The post-tax equity IRR for the project activity at the time of investment decision for the current project activity comes out to be 7.27%/17/.

The project activity is a renewable source of electricity generation and supplies the electricity to the Indian Electricity grid. The key parameters which determine the Equity IRR of the project activity are project cost, PLF and profitability estimates.

In the provided VCS PD/04/, the project cost is based on the Investment note

Particulars	Value	Unit	Assessment
Capacity of the project	100.00	MW_{AC}	The installed capacity of the project activity is 100 MW_{AC} which is verified from the letter of award (LOA)/22/ and further it was cross checked by the Investment note available at the time of decision making, further also verified from commissioning certificate/01/ and power purchase agreements/09/.
	140.00	MWp	
Project Life Time	25	Years	The operational lifetime of the project activity is sourced from Investment note which was available at the time of investment decision and further crosschecked with the technical data sheet/03/ provided by the project owner and further found in line with DPR/10/ value. This is also in accordance with the operating life established by the Central Electricity Regulatory Commission. Hence, the value for life time considered by project owner is correct and appropriate for the project.
Plant Load Factor	24.40	%	The PLF is considered as 24.60% which is based at estimated energy production calculations sourced from Investment note/37/ prepared by AMP Energy Green Seven Private Limited which was available at the time of investment decision. Further, also crosschecked from PVsyst report/24/ which found acceptable to the verifier's team. Hence the value considered by the project owner for demonstrating additionality of the project is deemed acceptable to the VVB team and also in line with paragraph 3 (b) of "Guidelines for the reporting and Validation of Plant Load Factors" (Annex 11 of EB 48). Hence the value considered by the project owner in the investment analysis is acceptable to the VVB team. Also, VVB team checked the actual PLF generation achieved by the solar plant for the recent operational years 2017 to 2022 and found that the average PLF achieved is approximately 18.48%, which is less than the figure (23.30%) achieved in sensitivity analysis with a 10% variation mentioned in IRR sheet/17/. VVB

			team carried out its own an independent assessment, which reveals that the project would become non-additional if PLF breaches the variation required to reach benchmark that is by +20.08%, which has the corresponding PLF value of 23.30% which is unlikely scenario. The VVB team thus considers that the Plant Load factor taken into account in the investment analysis is adequate and acceptable.
Second year degradation factor	0.30%	%	This value is sourced from Investment note which was available at the time of investment decision. Further, VVB team has cross verified with the NERL report on Photovoltaic Degradation Rates - An Analytical Review [35]. The report covers nearly 2000 degradation rates all across the globe and degradation rates has a mean of 0.8% per year. Also, normally most of the PV panels manufacturer[36] guaranteed 2-3% degradation by the first year and 0.7% on each year up to 10 years. Further, also referred manufacturer's datasheet of used panels for the same and found acceptable. So, the value considered in the investment analysis is acceptable to the VVB team, even total removal of the value does not render the project non -additional.
Annual degradation (3nd years onwards)	0.7		
Project cost	6,000.00	INR Million	The amount INR 6,000.00 million INR considered as the net project cost as a cash outflow in the Post tax equity IRR calculations. The project cost taken to demonstrate the additionality is based on the Investment note which was available at the time of investment decision to the project owner. However, as an additional check, the VVB team cross checked audited balance sheet/25/ incurred by the project owner for the project activity through Land Sale deeds and EPC orders/09/ placed to the major equipment suppliers/09/ evidence for the investment as per the requirements set forth by CDM VVS paragraph 99. VVB team confirmed that the investment as per EPC contract and service cost of INR 6,145.30 million is 2% higher than the estimated investment cost as per the DPR. With the benchmark at 7.27% decrease in the estimated project cost, the estimated cost is considered as conservative.

[35] https://www.nrel.gov/docs/fy12osti/51664.pdf

[36] https://www.solarquotes.com.au/blog/solar-panel-degradation/

			<table><tr><th>Particulars</th><th>Amount (in INR Million)</th></tr><tr><td>Civil Cost</td><td>90.20</td></tr><tr><td>EPC & Service Cost</td><td>4993.0</td></tr><tr><td>Supply Cost of inverter and BOP</td><td>81.11</td></tr><tr><td>Floters purchase and module cost</td><td>980.99</td></tr><tr><td>**Total Cost**</td><td>**6,145.30**</td></tr></table> Since there are additional costs also involved in the implementation of a solar power plant (include the broad heads like soft cost, IDC) that are not covered under the EPC contracts and service cost totalling to INR 6,145.30 million INR, CA is being referred to for cross verification of the actual capital expenditure of INR 6,145.30 million (the EPC cost and land cost sums up to the total expenditure). The actual investment of INR 6,145.30 million as per the CA certificate is 2% higher than the estimated cost. As the actual project cost or the cost of EPC and Land cost combined is less than the breakeven limit the verifier confirmed the robustness of the estimated project cost. As a consequence, when project cost from actual cost is compared with cost mentioned in Investment note, the post-tax equity IRR remains lower than the benchmark. A threshold analysis was carried out and found that the project would become non additional only if project cost goes down by -15.87%. However, reduction in project cost is not a likely scenario in the VVB team's opinion, as the project has been already commissioned and also actual cost incurred by the project owner is supported by the purchase orders/31/ and Chartered accountant audited balance sheet/10/.
Debt	75.0%	%	The debt equity ratio is based on the Investment note which was available at the time of investment decision. The actual financing pattern yields a gearing of 75:25 which is based on actual loan sanctioned/23/ to the project activity by the bank. Hence the debt equity ratio considered is acceptable.
Equity	25.0%	%	
Interest rate	9.15%	%	The interest rate is based on Investment note which was available at the time of investment decision. Also, Verifier's team cross verified the actual interest rate with debentures trust deed.

			The interest rate determined in Central Electricity Regulatory Commission Tariff order number SM/03/2016 (Suo-Motu) dated 30-March-2016[37], is 12.76%, which is higher than the interest rate considered in the IRR sheet/17/. However, Verifiers team even compared estimated interest rate with the actual rate and found there is no major impact on IRR and it is well below the benchmark. Hence accepted.
Debt Repayment tenure	20	Years	Loan Tenure is based on the Investment note which was available at the time of investment decision. The loan tenure suggested in the Central Electricity Regulatory Commission Tariff order number SM/03/2016 (Suo-Motu) dated 30-March-2016 is 13 years with 0-year moratorium and 13 years repayment. Hence the project considers conservative value in both moratorium period (01 years) and repayment period (15 years). VVB team also verified the debenture trust deed and found that the actual repayment period is 15 years. Thus, the repayment period considered is on par with the actual period. Hence, the repayment period & moratorium period considered for IRR calculation is found to be appropriate.
Moratorium	01	Years	
Operation and Maintenance	0.259	INR Million /MW	The O&M cost and its escalation is based on the Investment note which was available at the time of investment decision. The O&M cost suggested in the Central Electricity Regulatory Commission Tariff order number SM/03/2016 (Suo-Motu) dated 30-March-2016[38] is 0.259 INR million /MW. It is observed that O&M cost is not a critical factor at all, as 286.00% reduction in O&M cost (which in effect means free O&M service) would render the project non-additional. Further the -286.00% reduction in O&M cost is not a likely scenario in terms of project type and its context. The VVB team crosschecked the actual O&M cost from the O&M Commercial offer/28/ issued by Sterling & Wilson of the project activity i.e., 0.160 million INR, which is 61.77% lower than the values assumed during the investment decision making time, further crosschecked the escalation of O&M cost from O&M Commercial offer /28/ and found 4% which is same as it estimated value. Hence the
Escalation in O & M	4	%	

[37] sm_3.pdf

[38] https://cercind.gov.in/2016/orders/sm_3.pdf

			assumption of O&M cost and its escalation is acceptable to VVB team. After evaluating the project activity's values, it was determined that it was suitable for its location. Therefore, the VVB team accepts the assumption of O&M cost and its increase.
Insurance cost	8.85	Million	Insurance cost is considered as 0.0885 million/MW of the project cost each year and is sourced from the Investment note which was available at the time of investment decision. Further, the insurance cost is compared with the actual insurance policy opted by the project owner for project activity. The premium paid by the project owner is INR 140 Lakh per annum inclusive of 18% GST. Hence, the value of insurance charges is accepted.
Tariff	3.21	Rs/kWh	The tariff base rate is based on Investment note/37/, which was available at the time of investment decision. This is also crosschecked through the actual invoices/27/ raised to M.P. POWER MANAGEMENT COMPANY LIMITED (MPPMCL) and found to be similar as 3.21 Rs/kWh. The PPA/09/ is fixed without any escalation for 25 years. Hence, the tariff considered in the investment analysis is acceptable and found to be appropriate. Further increase in tariff is the unlikely scenario as the tariff is fixed without any escalation for 25 years from the commercial operation date. VVB team also verified the actual invoices/26/ raised by the project owner to MPPMCL and found the actual tariff is INR 3.21 /kWh. Since tariff is finalized during the bidding process hence didn't find appropriate to compare with the approved projects. Hence tariff rate considered in the investment analysis is deemed appropriate and acceptable to the VVB team.

Interest on working capital	9.15	%	The working capital requirements is based on Central Electricity Regulatory Commission[39] which was available at the time of investment decision. The working capital requirement for solar PV projects suggested in the Central Electricity Regulatory Commission Tariff order number SM/03/2016 (Suo-Motu) dated 30-March-2016 is Interest on working Capital is 13.26%, No of Days receivable is one month. Even with the interest rate of 13.26%, there is no major impact on IRR and it is below the benchmark. Also, the working capital has been added back to the cash inflow in calculation of the post-tax equity IRR of the project activity which is in line with paragraph 14 of the applied Tool 27/17/. Hence values considered in the investment analysis is conservative. Further during review of Credit Arrangement letter issued by IREDA (Indian Renewable Energy Development Authority with ref no. TS-11/12/2022-IREDA/1069, dated 11-November-2022/33/, which states that "the rate of interest for each working capital drawl of the facility will be stipulated by the bank at the time of disbursement of each drawl, which shall be sum of the Repo rate + Spread per annum, plus applicable statutory levy." Moreover, during interview with the project owner, no working capital has been taken from bank till date. Therefore, assessment team assessed independently through IREDA website. Hence interest rate on working capital and repayment tenor considered in the investment analysis is deemed appropriate and acceptable to the VVB team.
Depreciation Rate (Book)	4.00	%	The project owner has considered straight-line method for book depreciation where 90% of the initial value of the project cost is depreciated for the life period of the project considering 10% salvage value. This is as per as per Schedule XIV of the Companies Act, 1956 for computing book profit which is as per accounting practices followed in the host country. The following link has been verified and found correct.

[39] https://cercind.gov.in/2016/orders/sm_3.pdf

			https://taxguru.in/company-law/rates-depreciation-companies-act-2013.html
Residual Value	600	INR Million	The Residual Value is calculated in the IRR sheet/17/. The residual value is taken as 10% of the Depreciable cost in the project cost + Cost of EPC, which is in conformity with the best international practices and local accounting principles Also the same is in line with Salvage value provided in the Central Electricity Regulatory Commission Tariff order number SM/03/2016 (Suo-Motu) dated 30-March-2016[40] which was available at the time of investment decision/30/. Further VVB team cross checked from Section 205 (2b and c) of Companies Act 1956, which allows a depreciable cost of ninety five percent which implies a consideration of 10% of salvage value as a standard accounting practice. This can be verified from the below link https://taxguru.in/company-law/rates-depreciation-companies-act-2013.html As required by Tool 27/08/ the expected realisation on the sale of assets at the end of the operating life has been taken as residual value in the terminal year in the cash inflow in calculation of the post-tax equity IRR. The principle adopted conforms to the accepted accounting and taxation principles. Hence the salvage value considered in the project owner is appropriate and conservative.
IT Depreciation Rate	40	%	The IT depreciation is based on the https://incometaxindia.gov.in/charts%20%20tables/depreciation%20rates.htm available at the time of investment decision/30/. The project owner considered the IT depreciation rate 40% for power generating units. This is as per Income Tax Act 1961 stipulated for income tax calculation which is as per accounting practices followed in the host country. The following web link has been verified and found correct. https://incometaxindia.gov.in/charts%20%20tables/depreciation%20rates.htm

[40] https://cercind.gov.in/2016/orders/sm_3.pdf

Effective Income tax rate	33.38	%	The corporate tax payable is calculation based on the base corporate tax, Surcharge & educational cess. The calculation based on the following values: Income tax rate- 30% MAT – 15% Surcharge – 7% of corporate tax Educational Cess- 4% of corporate tax Service Charge – 18% The corporate tax value considered is correct and applicable to the project activity. The same has been verified in the following weblink and found to be correct. https://incometaxindia.gov.in/charts%20%20tables/depreciation%20rates.htm

Financial calculation and conclusion:

The Post tax equity IRR calculations were provided in a spreadsheet/11/. The calculation was verified and found to be correct by project VVB team; as well as the assumptions used in the calculation were deemed to be correct. The Post tax equity IRR without carbon credit revenues is;

Post tax Equity IRR	Benchmark Value
7.27%	14.97%

This demonstrates that the proposed project activity, in the absence of carbon credit advantages, is not financially advantageous when compared to the benchmark return on equity of 14.97%.

Sensitivity Analysis:

The guidance on assessment of investment analysis requires the robustness of the result reached to be demonstrated by a sensitivity analysis by altering the essential assumptions to a reasonable variation. The project owner has identified generation (PLF), project cost, O&M cost and tariff as critical assumptions. These constitute more than 20% of the project cost/revenue. As per para 28 of Tool 27 “Investment analysis”/17/ version 14.0 states that as a general point of departure, variations in the sensitivity analysis should at least cover a range of +10% and –10%, unless this is not deemed appropriate in the context of the specific project circumstances. Since project has already been implemented any variation in project cost is hypothetical. Nevertheless, the project cost has also been subjected to 10% variation. The sensitivity analysis reveals that excepting when the power tariff or PLF goes up by 10% or project cost comes down by 10% as given in the following table.

The results of sensitivity analysis show that even with a variation of ±10% in tariff, PLF, project cost, and O&M cost, Post Tax equity IRR is significantly lower than the benchmark. And it is evident from the results given above; the project remains additional even under the most favourable conditions. Also, the reasonable variations for these parameters were checked by calculating the variation necessary to reach the benchmark and then discussing the likelihood for that to happen.

Variation %	-10%	Normal	10%	Variation required to breach benchmark	Value required to breach Breaching
PLF (%)	3.79%	7.27%	10.91%	20.08%	29.03%
O&M Cost (INR (Mn))	7.47%	7.27%	7.02%	-347.00%	48.29
Project Cost (INR (Mn))	10.86%	7.27%	4.41%	-18.50%	4,890.00
Tariff (INR/kWh)	3.79%	7.27%	10.91%	20.08%	3.75

a) **Tariff:** The Investment Note/37/, which was accessible at the time of the investment decision/30/, is the source of the power tariff rate of 3.21 INR/kWh utilized for the investment analysis/11/. Additionally, the project will exceed the benchmark value when the tariff rate reaches 3.75 INR/kWh and at a tariff variation of 20.08%. A rise in tariff is unlikely because the power purchase agreement is signed for 25 years at a fixed rate of 3.21 INR/kWh.

b) **PLF:** The PLF value taken into consideration is based on Investment Note/37/ 24.40%, and at PLF variations of more than 29.03% and when PLF reaches 20.08% the IRR will exceed the benchmark amount. Since the PLF is determined using the expected yearly radiation assessment, it is improbable that the PLF value would grow to exceed the benchmark. Calculated from the actual generation records/35/, the real PLF from the project activity is below Calculated from the actual generation records/35/, the real PLF from the project activity is below the breaching value required. Further yearly PLF is listed as follows.

Financial Year	Electricity Generation (in KWh)	PLF
FY 06-24 to 09-24	29,378,614	11%

The PLF averages about 11% over the June 2024–September 2024 period, which is less than the 23.30% threshold needed to surpass the IRR benchmark figure.

c) **Project Cost:** The project cost for investment analysis is 6000 million INR. The cost is based on Investment Note/37/, which was accessible at the time of Investment Decision/30/. A deviation of -18.50% is necessary for IRR/11/ to exceed the benchmark, which is unlikely because the project has already been commissioned. The actual cost of project commissioning is 6,145.30 million INR (EPC cost)/31/ and 6,145.30 million INR (CA audited Balance Sheet), which is greater than the value necessary to exceed the benchmark while remaining within the sensitivity range applied.

d) **O&M Costs:** The O&M costs are based on the investment note, which was available at the time of the investment decision. Sensitivity analysis shows that O&M will exceed the benchmark at -347.00%, which is a hypothetical scenario. O&M expenses are vulnerable to escalation and inflationary pressure; therefore, any reduction is quite unlikely. The actual O&M was carried out at INR 160,000 Mn/MW, as per the audited balance sheet, with an equity IRR much lower than the benchmark figure.

Step 3: Barrier Analysis

The additionality of the project has been demonstrated by applying the investment analysis, thus no barrier analysis is carried out.

Step 4: Common Practice Analysis

The section below provides the analysis as per step 4 of the tool 01 “Tool for the demonstration and assessment of additionality”, version 7.0.0/08/ and according to tool 24 “Common Practice” Tool version 03.1/08/.

Step 1: Calculate applicable capacity or output range as +/- 50% of the total design capacity or output of the proposed project activity:

The bundled project installed capacity is 100 MW. Therefore, total capacity of power plants which will be included in the analysis will be between 50 MW to 150 MW.

Step 2: Identify similar projects (both CDM and non-CDM) which fulfil all of the following conditions:

Identify similar projects (both CDM and non-CDM) which fulfil all of the following conditions:

(a) The projects are located in the applicable geographical area;

(b) The projects apply the same measure as the proposed project activity;

(c) The projects use the same energy source/fuel and feedstock as the proposed project activity, if a technology switch measure is implemented by the proposed project activity;

(d) The plants in which the projects are implemented produce goods or services with comparable quality, properties and applications areas (e.g., clinker) as the proposed project plant;

(e) The capacity or output of the projects is within the applicable capacity or output range calculated in Step 1;

(f) The projects started commercial operation before the VCS PD is published for public comment or before the start date of proposed project activity, whichever is earlier for the proposed project activity.

Identification of the similar projects (CDM and non-CDM) is carried out as per sub-steps of Step (2) as follows:

VVB team noted that the applicable geographical area is the entire host country i.e., India however project owner has chosen the geographical area as Madhya Pradesh state in India for analysis where the project is located. Despite the fact that the bids for the projects were conducted by RUMSL/38/ in accordance with the criteria given by the Ministry of Power, the Government of India, this project activity-related bid was state specific i.e., state of Madhya Pradesh.

VVB team confirms during the site visit that the energy source used by the project activity is solar. Hence, only solar energy projects have been considered for analysis.

Since the project activity is 100 MW, the output range of +/- 50% has been considered as 150 MW (Higher range for comparison) to 50 MW (Lower range for Comparison) which is assessed to be correct.

Purchase Order/Purchase contract date of the project activity is 28-January-2016, therefore, projects, which have started commercial operation before 28-January-2016, have been considered for analysis.

Numbers of Similar projects identified which fulfil above-mentioned conditioned are

N_{solar}= 0

VVB noted that all the above information is correct and is sourced from government of India databased and hence acceptable.

Step 3: Within the projects identified in Step 2, identify those that are neither registered CDM project activities, project activities submitted for registration, nor project activities undergoing verification. Note their number, N_{all}.

Out of N_{solar} projects, neither registered CDM project activities, project activities submitted for registration, nor project activities undergoing validation. Note their number.

Hence N_{all} = 0[41]

The assessment team reviewed the CERC guidelines and confirmed that no project activities with a similar design in the specified geographical location have been submitted for registration or are currently undergoing validation.**Step 4:** within similar projects identified in Step 3, identify those that apply technologies that are different to the technology applied in the proposed

[41] Microsoft Word - CERC RE-Tariff-Regualtions 2017-20

project activity. Note their number as N_{diff}. As per the tool 24 Common Practice/08/ version 3.1., the project activities have been separated from the different technologies on the basis two criteria:

Size of Installation - VVB team confirms that as the proposed project activity is a large-scale project and applies large scale CDM methodology i.e., ACM0002 version 22.0/08/.

Investment climate on the date of the investment decision as per paragraph 12 d) of the Tool 24 "Common practice" version 3.1/08/. – The project proponent has considered the projects implemented in different investment climate and under different legal regulations as different technology.

Hence, projects either of the conditions is satisfied those projects are counted for calculating N_{diff} projects.

Thus, $N_{diff} = 0$

The assessment team reviewed the CERC guidelines and confirmed that no similar projects utilizing technologies different from those proposed in the activity were identified.**Step 5:** calculate factor $F = 1 - (N_{diff}/N_{all})$ representing the share of similar projects (penetration rate of the measure/technology) using a measure/technology similar to the measure/technology used in the proposed project activity that deliver the same output or capacity as the proposed project activity.

The factor F was found to be in line with Tool 24 "Common practice" version 3.1/16/.

$F=1- (N_{diff}/N_{all}) = 1- (N_{all}/N_{all}) = 1- 1 = 0$. $N_{all} - N_{diff} = 0$

As per methodological tool "Common practice" version 03.1, the proposed project activity is a "common practice" within a sector in the applicable geographical area if the factor F is greater than 0.2 and $N_{all} - N_{diff}$ is greater than 3. Thus, if both conditions are fulfilled, then project activity will be a common practice. Otherwise, the project activity is treated as not a common practice.

Outcome of Step 5:

Thus, the project does not fulfil the requirement that F is greater than 0.2 and $N_{all}-N_{diff}$ is greater than 3, and the project is not a common practice.

The above discussions show that solar power development is not a financially attractive; hence the project activity is additional and the assessment team considers the approach and calculations acceptable as per the requirements in the methodological tool.

CL 01 & CAR 02 were raised on this section and closed successfully. Please refer Appendix 3 for further details.

3.3.6 Quantification of GHG Emission Reductions and Carbon Dioxide Removals

Assessment team checked the baseline, project and leakage calculation and confirm that the evaluation of baseline, project and leakage is as per the approved methodology and formula used to calculate the same is correct. The detail analysis is as below:

Baseline Emissions: As per para 57 of ACM0002: Grid-connected electricity generation from renewable sources - Version 22.0, the Baseline emissions for other systems are the product of amount electricity displaced with the electricity produced by the renewable generating unit and an emission factor.

$BE_y = EG_{PL,y} \times EF_{CO2,y}$

where:

BE_y = Baseline emissions in year y (tCO_2/yr)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EG_{Facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

$EF_{gridcm,y}$ = CO_2 emission factor for the grids electricity in year y (tCO_2/MWh)

BE_y = 211,556 x 0.9352 = 197,846 tCO_2 (Round-down Annual average value)

Project Emissions: -

As per para 40 of the applied methodology ACM0002 (version 22.0), for most renewable energy power generation project activities, PEy = 0

Since, this is a renewable energy power project, the project emissions are considered as zero.

Hence, project emissions **PEy = 0 tCO2e**

Leakage Emissions: -

As per Para 71 of ACM0002 Version 22.0 The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g. extraction, processing, transport etc. are neglected. No other leakage emissions are considered.

The net emission reduction is calculated as

$ER_y = BE_y - PE_y$

Vintage period	Estimated baseline emissions (tCO_2e)	Estimated project emissions (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated reductions VCUs (tCO_2e)	Estimated removals VCUs (tCO_2e)	Estimated total VCUs (tCO_2e)
06-June-2024 to 31-Dec-2024	114,459	*0*	*0*	114,459	*0*	114,459
01-Jan-2025 to 31-Dec-2025	199,292	*0*	*0*	199,292	*0*	199,292
01-Jan-2026 to 31-Dec-2026	198,694	0	0	198,694	0	198,694
01-Jan-2027 to 31-Dec-2027	198,097	0	0	198,097	0	198,097
01-Jan-2028 to 31-Dec-2028	197,503	0	0	197,503	0	197,503
01-Jan-2029 to 31-Dec-2029	196,910	0	0	196,910	0	196,910
01-Jan-2030 to 31-Dec-2030	196,319	0	0	196,319	0	196,319
01-Jan-2031 to 05-June-2031	83,653	0	0	83,653	0	83,653
Total	**1,384,927**	**0**	**0**	**1,384,927**	**0**	**1,384,927**

Parameters determined ex-ante:

Baseline emission factor of Indian Grid is established ex-ante based on Tool to calculate the grid emission factor, using a combined approach consisting 75 % operating margin and 25 % build margin. The emission coefficient from official data published in Central Electricity Authority (CEA) CO2 Baseline database available to the project Proponent at the time of submission of joint VCS PD & MR for joint validation & Verification and global stakeholder's consultation process. CEA is an official source of Ministry of Power; Government of India have worked out baseline as CO2 baseline database. The assumption was verified by the validation team and found to be correct.

$EF_{grid,OM,y}$:- Operating Margin CO_2 emission factor in year

$EF_{grid,OM,y}$ = 0.9580 tCO_2e/MWh

VVB team confirmed that, Operating margin CO_2 emission factor for the project electricity system in year y Calculated as the last 3-year (2020-21, 2021-22 and 2022-23) generation-weighted average, sourced from Baseline CO_2 Emission Database, Version 19.0, November-23 published by Central Electricity Authority (CEA), Government of India.

$EF_{grid,BM,y}$:- Build Margin CO_2 emission factor in year y.

$EF_{grid,BM,y}$ = 0.8670 tCO_2e/MWh

VVB team confirms that, build margin CO_2 emission factor for the project electricity system in year y Baseline CO_2 Emission Database, Version 19.0, November-23 published by Central Electricity Authority (CEA), Government of India.

$EF_{grid,BM,y}$:- Combined Margin CO_2 emission factor in year y

$EF_{grid,CM,y}$ = 0.9352 tCO_2e/MWh

VVB team confirms that, build margin CO_2 emission factor for the project electricity system in year y Baseline CO_2 Emission Database, Version 19.0, November-23 published by Central Electricity Authority (CEA), Government of India.

Combined margin CO_2 emission factor for the project electricity system in year y Calculated as the weighted average of the operating margin (0.75) & build margin (0.25) values, sourced from Baseline CO_2 Emission Database, Version 19.0, November-23 published by Central Electricity Authority (CEA), Government of India.

Parameters determined ex-post:

The project applies approved consolidated methodology ACM0002: Grid-connected electricity generation from renewable sources --- Version 22.0, as per the applied methodology "Quantity of net electricity displaced as a result of the implementation of the CDM project activity in year y (MWh)" i.e.

$EG_{Facility,y}$ is to be monitored. For each project instances the value for the all the project instances is given below and the value is the estimated value for the project activity, Project activity generated 211,556 MWh amount of electricity in year, and it will generate 1,480,892 MWh amount of electricity in a crediting period.

The parameters monitored ex-post involves quantity of net electricity displaced as a result of the implementation of the project activity. As per the final VCS PD, Joint Energy Meter Reading Report will be the source of data during verification. The VVB will use the same source for verification of emission reductions. In accordance with the methodology requirement, net electricity supplied by the project activity is obtained from Joint Energy Meter Reading Report issued by State electricity authority which provide input values used for calculation of $EG_{Facility,y}$ by the project activity and forms the basis for emission reduction calculation. Electricity export to the grid is measured by energy meters. The main meter reading is taken jointly on a fixed day of every month for the preceding month at the delivery point and signed by the representatives of project developer and customer.

The calculation of emission reductions is based on the net power delivered to the grid, as recorded in the Monthly Joint Meter Reading Report. This figure can be verified against the consumer's invoice.

The floating solar PV plant's electricity has been transmitted to the utility substation at Saktapur's 220 kV substation. The state utility, MPPMCL, provide the project owner with Monthly Generation Reports based on the amount of power transmitted into the substation as shown by the electrical meters at the supply feeding point. Project activity also has its own main and check meters at the pooling substation. Transmission losses for each project are estimated by comparing bulk meter readings to the specific project meters. The net electricity delivered is calculated by subtracting the import value from the export value.

Each month, representatives from both the project owner and DISCOM verify the readings from the main and check meters at the metering site.

State VVB team confirms that all the energy meters (main meter) installed at each building/block are of accuracy class of 0.2s and are calibrated as per the calibration frequency mentioned in monitoring plan in VCS PD i.e. The calibration frequency of meters is once in 5 years. The meters at the project site were calibrated as per guideline of the central electricity authority (CEA) & same has been confirmed through the document review. Also, the meters at the project site were tested by the electricity transmission department. VVB team confirm that, the export and import electricity by the project activity. Data will also be recorded by SCADA system. The calibration frequency of the monitoring meter is once in a five-year, PP has used secure meter (Energy meter), meters were connected at Interconnection point (Export switchgear). The details of the monitoring meter have been provided in appendix 6 of the validation report. VVB Team conclude that all project relevant assumptions and data mentioned in project description along with their reference and sources are found correct and reasonable in the context of the project activity. Further, VVB confirmed that project activity is in conformance with the VCS program rules. Calculation regarding baseline emission is found correct and can be replicate using data and parameter values provided in the project description. In normal conditions, the main meter readings are used for accounting; however, if the main meter fails, the check meter readings are used instead. Electricity meters are calibrated according to national standards, which require recalibration at least every five years or if there are any inconsistencies between the main and check meters. Monitoring of this parameter is recorded monthly for emission reduction purposes, and electricity generation values are calculated following the method in Section 5.3 of the Project Design (PD). The cross-checking mechanism is also consistent with the procedures in this section.

3.3.7 Methodology Deviations

Assessment team confirms that no methodology deviation is applicable for the present project activity.

3.3.8 Monitoring Plan

Assessment team checked the monitoring practice onsite and checked the guideline of respective State Electricity authority of the concerned. The detail analysis is as below:

Parameters determined ex-ante:

As per the approved methodology ACM0002: Grid-connected electricity generation from renewable sources (Version 22) baseline emissions for the project activity are the product of electrical energy baseline $EG_{Facility,y}$ expressed in MWh of electricity produced by the renewable energy generating unit multiplied by the grid emission factor

$BE_y = EG_{PJ,y} \times EF_{CO2,y}$ [Equation (1)] of applied methodology

where:

BE_y = Baseline emissions in year y (tCO2/yr)

$EG_{PJ,Y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EF_{Grid,CM,y}$ = Combined margin CO2 emission factor for grid connected power generation in year y calculated using TOOL07 (tCO2e/MWh)

Data and Parameters	Unit	Ex-ante Determined Value
$EF_{grid,OM,y}$	tCO_2e/MWh	0.9580
$EF_{grid,BM,y}$	tCO_2e/MWh	0.8670
$EF_{grid,CM,y}$	tCO_2e/MWh	0.9352

The parameters monitored ex-post:

Parameters	Description	Measurement method and QA/QC procedures	Assessment conclusion
$EG_{Facility,y}$	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr) quantity of net electricity displaced as a result	The electricity generated by solar power plant is measured by parameter $EG_{facility,y}$ measured with a set of meter located at four set of meters located at the NTPC substation located at sub-station, further generated electricity transfer to MPPTCL 220 kV Saktapur sub-station. Net electricity generation amount (exports – imports) is	Consistent with methodology

Parameters	Description	Measurement method and QA/QC procedures	Assessment conclusion
	of the implementation of the VCS project activity in year y (MWh).	summarized on the meter or if meter readings are conducted for exports and imports separately. Net electricity supplied to the grid is monitored continuously by ABT bidirectional tri-vector, main and check meters located at substation. These meters maintained by the technology supplier as per the operation and maintenance contract with the PP. Gross electricity generated by the project activity of solar power plant & same is verified with monthly generation records. provided by PP. The Assessment team checked the monthly figures and compared to the calibration certificate of the plant operator. The meters provide the net generation figures, i.e. gross electricity minus imports. The data is automatically transferred to the data control system online. The system can show different kind of data aggregation levels. Assessment team also cross checked the net electricity exported to grid values with export data and found consistent. During the current monitoring period, Assessment team checked the calibration compliance of the main and check meters and found that, All the meters are pre-calibrated and no delay in schedule calibration found. The data will be cross checked with sales receipts.	

Monitoring Arrangement:

The net generated electricity from the solar project is measured using an energy meter located at the solar project site. Then, the SCADA system is used for managing and monitoring the project operations, including the data collection from field equipment, the data dashboard, and execution of control orders to site officer. The site officer summarizes the data on daily basis, and prepares Electricity Meter Reading Data Report to the electricity buyer on monthly basis.

Organizational structure and responsibilities:

The responsibilities and authorities of project management, data handling and recording, measurement methods and QA/QC procedure have been systematically established and formalized and the same was verified during the online visit.

Data Measurement & Data collection and archiving:

The export and import energy or net electricity displaced by project will be measured continuously using above mentioned energy meter located after inverters. Readings of meters shall be taken on monthly basis by authorized officer of developer in the presence of PP or representative of PP. In the event of failure of generation meter, the user energy meter or inverter generation value will be used in monitoring the electricity generation data. Readings from meters will be collected in the presence of the plant in-charge. Export and import or net electricity displaced by project would be recorded and stored in logs as well as in electronic form on a daily basis. The records are checked periodically by the Plant Manager and discussed thoroughly with the plant supervisor. The period of storage of the monitored data will be 2 years after the end of crediting period or till the last issuance of VERs for the project activity whichever occurs later.

QA/QC Procedures:

The energy meters are installed at the respective project site. The calibration of all the meters will be undertaken at required intervals (at least once in five years as per CEA notification) and faulty meters will be duly replaced immediately. Records of calibration certificates will be maintained for verification purposes. Hence, a reliable method will be ensured with monitoring of the parameters. The invoice records will be used and kept for cross checking the consistency of the recorded data.

Internal Audit Procedure and Procedure for handling non-Conformances with Validated Monitoring Plan

The O&M and site officer will monitor the electricity generation via SCADA system. The site manager consistently prepares and updates net electricity generation reports. This report is reviewed monthly by Site Manager. The monthly records will be further checked by VCS Project Team and will calculate GHG emission reduction from the solar project. It will be ensured that adherence to monitoring parameters will be maintained and identify non-conformance if any. In the event of non-conformances being detected, prompt corrective actions are taken to address and resolved appropriately.

Solar Panel Cleaning Procedure

The cleaning procedure for the solar panels is performed manually, and they are cleaned at least monthly by operation and maintenance team of the project proponent. The frequency of cleaning has been stated in the O&M agreement. There is no soap involve in the cleaning process.

Apportioning: -

In case the start and end dates of a particular monitoring period do not match with the dates of the billing period, the net electricity exported to the grid would be calculated from the difference of number of days which are not matching of billing period and monitoring period or apportioned in line to the daily generation values for the said mismatch period.

The calculated value after apportioning would be used for calculation of emission reductions during that period.

Personnel training:

In order to ensure a proper functioning of the project activity and a properly monitoring of emission reductions, the staff will be trained. The plant helpers will be trained in equipment operation, data recording, reports writing, operation and maintenance and emergency procedures in compliance with the monitoring plan. The validation team confirmed the above by reviewing the monitoring manual set by PP and confirmed the implementation of the training through the training records.

Based on the on-site visit and interviewed with the site in-charge and project proponent representative Manager of AMP Energy Green Seven Private Limited the VVB team confirmed that monitoring parameter has been correctly descried in the updated PD and in compliance with the methodology ACM0002 version 22.0 and the guidance given by VCS Standard version 4.7 and VCS program guide version 4.4.

CAR 02 was raised in this section and closed successfully. Kindly refer appendix 3 for more information.

3.4 Non-Permanence Risk Analysis

The project activity does not require a non-performance risk analysis. Hence, this is not applicable.

4 VALIDATION OPINION

4.1 Validation Summary

VVB team confirms that, GHG statement is the responsibility of the project proponent. The project activity fulfils the criteria for the validation as established in section 4 of the VCS standard version 4.7. VVB team has reviewed the project description documents and subsequently carried out the onsite audit and follow-up interview with project proponent and confirm the fulfilment of stated criteria. VVB team has the unmodified opinion regarding the project activity, VVB team has checked all the details and found it consistent and free of material errors or misstatements.

As per VCS Standard version 4.7 Para 4.1.10, VVB team verified that, the criteria for validation of the project activity are in line with the relevant VCS Version 4 criteria and all the relevant host country (India) criteria. The objective of this validation activity is to have an independent third party for the assessment of the project design, estimated ER sheet and to ensure a thorough assessment of the proposed project activity against the applicable regulatory requirements.

VVB concluded the assessment, as per para 4.1.18 of the VCS standard version 4.7, with a positive validation opinion regarding the project activity. The VVB team has raised 1 Clarification (CL), 2 CAR (Corrective action request), and 00 FAR (forward action request). All the deviations identified as CARs and/or CLs have been successfully closed and mentioned in Appendix 3 of the validation report. The VVB team clearly state that there is no material misstatement at the level of the GHG statement and GHG declarations made by the Project Proponent.

4.2 Validation Conclusion

Our Validation approach was based on the requirements as defined under the applicable regulatory requirements. The validation has been conducted in accordance with the VCS Standard version 4.7, other valid and applicable VCS Regulatory Documents and templates (see Appendix 2 of this Validation Report below), UNFCCC rules and requirements (where applicable) and associated decisions of the VCS Secretariat and the UNFCCC Executive Board (where applicable). Our approach is risk-based, drawing on an understanding of the risks associated with estimated GHG emissions data and the controls in place to mitigate these. The validation can confirm that:

- VVB team verified that, the data & the information provided in GHG statement assertion is projected. All the assumptions are based on the available supporting document & third-party reports and/or those mentioned in appendix 2. The estimates are based on assumptions that are subject to change. Thus, VVB concludes that the project is likely to achieve estimated emission reductions as declared by the Project Proponent.
- The project's baseline and additionality are assessed against "ACM0002. Grid-connected electricity generation from renewable sources --- Version 22.0. and TOOL 01: Tool for the demonstration and assessment of additionality version 07.0.0.
- The project's monitoring plan is assessed against "ACM0002. Grid-connected electricity generation from renewable sources --- Version 22.0".
- A risk-based approach has been followed to perform this validation activity. The review of the project description and additional documents related to baseline and monitoring methodology; the subsequent background investigation, follow-up interviews with Project Proponent have provided LGAI Technological Center S.A. (Applus+ Certification) with sufficient evidence for positive validation opinion as per the requirement of VCS.
- Based on the assessment of the data, assumptions, limitations, and methods, a conclusion can be drawn regarding the likelihood of the proposed solar power project achieving the estimated GHG emission reductions.

- Thus, VVB concludes that the project is likely to achieve estimated emission reductions as described below:

Crediting period: From [06-June-2024] to [05-June-2031]

Validated estimated GHG emission reductions and carbon dioxide removals for the project crediting period:

Vintage period	Estimated baseline emissions (tCO_2e)	Estimated project emissions (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated reductions VCUs (tCO_2e)	Estimated removals VCUs (tCO_2e)	Estimated total VCUs(tCO_2e)
06-June-2024 to 31-Dec-2024	114,459	0	0	114,459	0	114,459
01-Jan-2025 to 31-Dec-2025	199,292	0	0	199,292	0	199,292
01-Jan-2026 to 31-Dec-2026	198,694	0	0	198,694	0	198,694
01-Jan-2027 to 31-Dec-2027	198,097	0	0	198,097	0	198,097
01-Jan-2028 to 31-Dec-2028	197,503	0	0	197,503	0	197,503
01-Jan-2029 to 31-Dec-2029	196,910	0	0	196,910	0	196,910
01-Jan-2030 to 31-Dec-2030	196,319	0	0	196,319	0	196,319
01-Jan-2031 to 05-June-2031	83,653		0	83,653	0	83,653

Vintage period	Estimated baseline emissions (tCO_2e)	Estimated project emissions (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated reductions VCUs (tCO_2e)	Estimated removals VCUs (tCO_2e)	Estimated total VCUs(tCO_2e)
Total	1,384,927	0	0	1,384,927	0	1,384,927

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Section	Information	Justification	Assessment method and conclusion
N/A	N/A	N/A	Not applicable as there is no commercial sensitive information available

APPENDIX 2: DOCUMENTS REVIEW UNDER VALIDATION

No.	Author	Title	References to the document	Provider
1.	REWA ULTRA MEGA SOLAR LTD. (RUMS)	Commissioning certificate	Issued Dated: 06-June-2024	Project Proponent
2.	Applus	Contract of the PP with the DOE	Contract of the PP with the DOE	Project Proponent
3.	NA	Technical specifications	Technical specifications of all equipment from Manufacturers brochure.	Project Proponent
4.	PP	VCS PD Version 01 VCS PD version 02 VCS PD Version 03 VCS PD Version 04 VCS PD Version 04 Final VCS PD Version 05	12-April-2024 08-July-2024 09-October-2024 16-October-2024 25-November-2024 25-June-2025	Project Proponent
5.	PP	Emission Calculation Sheet- version 1.0- Estimated ER Sheet V.02 Estimated ER Sheet V.03 Estimated ER Sheet V.04 Final Estimated ER Sheet V.05	12-April-2024 08-July-2024 09-July-2024 16-October-2024 25-June-2025	Project Proponent
6.	PP	• Validation and Verification Manual, Version 03.2 • Kyoto Protocol (1997)	https://cdm.unfccc.int/Reference/Standards/index.html https://unfccc.int/process-and-meetings/the-kyoto-protocol/what-	Project Proponent

No.	Author	Title	References to the document	Provider
			is-the-kyoto-protocol/kyoto-protocol-targets-for-the-first-commitment-period	
7.	NA	Single Line Diagram	Plant Single Line Diagram	Project Proponent
8.	UNFCCC VERRA	Tools/ guidelines used in the project activity	UNFCCC CDM web site • UNFCCC Methodology: ACM0002: Grid-connected electricity generation from renewable sources - Version 22.0 EB 122 Annex 2_ACM0002_ver22.0 (unfccc.int) • Tool 01 - Tool for the demonstration and assessment of additionality, Version 07.0.0 • Tool 07 - Tool to calculate the emission factor for an electricity system -Version 07.0 • TOOL05: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation version 03.0 • TOOL24: Common practice version 03.1 • TOOL27: Investment Analysis version 14.0 • Glossary of CDM terms version 11.0 • VCS standard V4.7 • VCS Program Guide V4.4 • Guidelines for the reporting and Validation of Plant Load Factors" (Annex 11 of EB 48)[42]	
09.	PP	• PPA • EPC Agreement	Power purchase agreement signed between AMP Energy Green Seven Private Limited and Consumer and Madhya Pradesh Power Management Company Limited EPC agreement signed between AMP Energy Green Private Limited and Floatex Solar Pvt. Ltd. & Arken Solutions Pvt. Ltd.	PP

[42] Microsoft Word - EB48_repan11_Guidelines on plant load factor.doc (unfccc.int)

No.	Author	Title	References to the document	Provider
		• Module supply Agreement • Central Inverter	1. Between PP and Vina solar technology co. ltd. 2. PP and Zhejiang Jinko solar co. ltd. PP and SUNGROW INDIA PRIVATE LIMITED	
10.	PP	• Detailed Project Report • AMP Energy Green Seven Private Limited Information Memorandum • Declaration of Annual O_M	Prepared by AMP ENERGY GREEN SEVEN PVT. LTD. Internal document prepared by PP AMP Energy Green Seven Private Limited	PP
11.	PP	LSC Document	Local stakeholders' consultation supporting; Notices, MoM's & photographs Feedback register etc.	Project Proponent
13.	PP	Declaration letter	For no double counting and part into any emission trading program and not received under another registry	Project Proponent
14.	PP	Local Stakeholder consultation	Snapshots and supportive documents, Feedback Forms, Meeting announcement documents etc.	Project Proponent
15.	PP	Module supply agreement Agreement	Ref: 1.Znshine 2. JA Solar 3. Trina Solar 4. Canadian Solar 5. Longi 6. Jinko	Project Proponent
16.	PP	Indian Sources/Regulatory links	• Electricity Act 2003[43] • National Electricity Policy 2005[44] • Tariff Policy 2006[45]	Project Proponent

[43] https://cercind.gov.in/Act-with-amendment.pdf

[44] https://indiatransmission.org/listdocuments/Policy/NEP?doc_id=1101

[45] https://cea.nic.in/wp-content/uploads/legal_affairs/2020/09/Tariff%20policy.pdf

No.	Author	Title	References to the document	Provider
17.	PP	IRR spreadsheet, V1.0 IRR spreadsheet, V2.0 IRR spreadsheet, V3.0 IRR spreadsheet, V4.0 IRR spreadsheet, V5.0 Final IRR spreadsheet, V6.0	12-April-2024 08-July-2024 09-October-2024 16-October-2024 25-November-2024 Final IRR spreadsheet, V5.0	Project Proponent
18.	PP	BM (Build Margin) OM (Operating Margin) _ calculation Sheet Measured Efficiency (Confidential)	-	Project Proponent
19.	PP	Invertor Supply	Ref: - 1.Sungrow	Project Proponent
20.	TUV SUD South Asia (India)	ESIA Report	May 2022	Project Proponent
21.	PP	Mega Solar Limited (RUMS) (RfP dated 02-November-2021	RfP dated 02-November-2021	Publicly available
22.	PP	Aswath Damodaran	-	Publicly available
23.	PP	Loan Sanction letter Issued by IREDA	Ref No. TS-11/12/2022-IREDA/1069 Dated- 11-November-2022	Project Proponent
24.	TUV Sud south asia	Third party PV Systemreport	Dated 09-January-2023	Project Proponent
25.	CA Certificate	Walkar Chandok	-	Project Proponent
26	PP	JMR	MPPMCL	Project Proponent
27	PP	Invoices	MPPMCL	Project Proponent
28	Sterling & Wilson	Commercial Offer	Dated 02/02/2023	Project Proponent

APPENDIX 3: CLARIFICATION REQUESTS, CORRECTIVE ACTION REQUESTS, FORWARD ACTION REQUESTS (CAR/CL/FAR)

Table 1.Remaining FAR from validation and/or previous verification

FAR ID	01	Section no.		Date : DD/MM/YYYY
Description of FAR				
N/A				
Project participant response				**Date :** DD/MM/YYYY
Documentation provided by project participant				
DOE assessment				**Date:** DD/MM/YYYY

Table 2. CL from this verification

CL ID	01	Section no.	3.1	**Date:** 01-April-2024
Description of CL				
Project Proponent shall submit the following documents to VVB team: - 1. Commissioning certificate of the project activity. 2. No double counting declaration along with declaration of intended use of Verified Carbon Units (VCUs) 3. No Double Claiming declaration with Emissions Trading Programs 4. Detailed project report 5. Third party report for PLF/CUF considered in the project activity 6. Meter photographs 7. Single Line Diagram (SLD) of the project activity 8. EPC agreement (If any) 9. Loan Sanction letter 10. Purchase orders or equivalent document for solar plant 11. Letter for Synchronization/Interconnection with DISCOM (If any) 12. Employment Records 13. Training Records 14. HR Policy				

15. Local Stakeholder consultation documents including photographs, invitation letters, minutes of meeting, attendance etc. 16. Major Breakdown details (if any) **1) Thus, CL will be open till submission of above requested documents.**	
Project participant response	**Date:** 08-July-2024
PP attached all the documents as required for VVB perusal	
Documentation provided by project participant	
1. Commissioning certificate of the project activity - Attached 2. No double counting declaration along with declaration of intended use of Verified Carbon Units (VCUs) - Attached 3. No Double Claiming declaration with Emissions Trading Programs - Attached 4. Detailed project report - Attached 5. Third party report for PLF/CUF considered in the project activity - Attached 6. Meter photographs - Attached 7. Single Line Diagram (SLD) of the project activity - Attached 8. EPC agreement (If any) - Attached 9. Loan Sanction letter - Attached 10. Purchase orders or equivalent document for solar plant - Attached 11. Letter for Synchronization/Interconnection with DISCOM (If any) – PP waiting for the same from the respective authorities 12. Employment Records - Attached 13. Training Records - Attached 14. HR Policy - Attached 15. Local Stakeholder consultation documents including photographs, invitation letters, minutes of meeting, attendance etc - Attached 16. Major Breakdown details – Attached	
VVB assessment	**Date:** 22-July-2024
Verifier found that all the supporting documents required for the validation of PD as mentioned in this CL has not been provided. Out of 16 required documents, only 8 documents have been made available and those are Commissioning certificate, No double counting declaration along with declaration of intended use of Verified Carbon Units (VCUs), No Double Claiming declaration with Emissions Trading Programs, Third party report for PLF/CUF, Meter photographs, Single Line Diagram (SLD) of the project activity, Loan Sanction letter and Purchase orders or equivalent document for solar plant. Rest documents have not been made available. Hence, CL is open.	
Project participant response	**Date:** 09-October-2024
PP attached all the pending documents as required for VVB perusal	
Documentation provided by project participant	
4. Investment Decision Note - Attached 8. EPC agreement (If any) – Not applicable & all the purchase orders have been provided to validator 11. Letter for Synchronization/Interconnection with DISCOM (If any) – PP waiting for the same from the respective authorities 12. Employment Records - Attached 13. Training Records - Attached 14. HR Policy - Attached	

<table>
<tr><td colspan="2">15. Local Stakeholder consultation documents including photographs, invitation letters, minutes of meeting, attendance etc – Attached
16. Major Breakdown details – Attached</td></tr>
<tr><td>VVB assessment</td><td>Date: 14-October-2024</td></tr>
<tr><td colspan="2">VVB team verified that; PP has submitted the following document:
1. PP has submitted the investment decision note of the project activity, same is found consistent.
2. PP and Arken Solutions Pvt. Ltd. have signed an EPC contract for 140 MWp (DC)/MW (AC). The signed contract was submitted to the VVB team, that are accepted it since it's consistent.
3. VVB team during document review found that; submitted copy of the letter of synchronization is found missing. Thus, finding is open.
4. Name of the technical training is found missing thus findings is still open.
5. PP has submitted the name and list of the employee, moreover to back the statement salary slip and employment contract is found missing, thus finding is open.
6. VVB team verified that; PP has submitted the HR policy of the company and it is find consistent and thus accepted.
7. VVB team verified that; PP has submitted the local stakeholder consultation details and VVB team has crosschecked the details with supporting documents i.e. invitation letter, meeting evidence, photos and it is found consistent. Thus, accepted.
8. PP has submitted the breakdown log sheet to the assessment team and it is found consistent thus accepted.</td></tr>
<tr><td>Project participant response</td><td>Date: 16-October-2024</td></tr>
<tr><td colspan="2">3. Letter for Synchronization is now provided to assessment team.
4. name of technical training is now provided
5. salary slip is now provided to assessment team.</td></tr>
<tr><td colspan="2">Documentation provided by project participant</td></tr>
<tr><td colspan="2">1. Letter for Synchronization/Interconnection with DISCOM document.
2. Technical training document,
3. Salary slip document.</td></tr>
<tr><td>VVB assessment</td><td>Date: 07-November-2024</td></tr>
<tr><td colspan="2">1. VVB team verified that, Letter of synchronization and letter of incorporation dated 07/06/2024 and it is find consistent and thus accepted.
2. VVB team verified that; PP has updated the technical training in project description, moreover to verify the same PP has submitted the training details document and found it consistent and thus accepted.
3. VVB team verified that; PP has submitted the salary slip and employment agreement and it is found consistent and thus accepted.
Thus, CL 01 is closed.</td></tr>
</table>

Table 3. CAR from this verification

<table>
<tr><td>CAR ID</td><td>01</td><td>Section no.</td><td>3.1</td><td>Date: 01-April-2024</td></tr>
<tr><td colspan="5">Description of CAR</td></tr>
<tr><td colspan="5">PP requested to revise the VCS PD as per the VCS-PD Template version 4.3 and as per the requirement of the VCS standard version 4.6.
Thus, Corrective action sought.</td></tr>
</table>

Project participant response	**Date:** 08-July-2024
PP revised the VCS PD as per the VCS-PD Template version 4.3 and as per the requirement of the VCS standard version 4.6.	
Documentation provided by project participant	
Revised VCS PD	
VVB assessment	**Date:** 22-July-2024
Verifier found that the version for VCS PD has been revised and has been made consistent to VCS PD v4.3 template. However, the version of VCS standard mentioned in the PD is 4.7 while the VCS Standard version mentioned in PP response is 4.6. Therefore, PP to clarify. Hence, CAR is open.	
Project participant response	**Date:** 09-October-2024
PP revised the VCS PD as per the VCS-PD Template version 4.3 and as per the requirement of the VCS standard version 4.7.	
Documentation provided by project participant	
Revised VCS PD	
VVB assessment	**Date:** 14-October-2024
VVB team verified that; PP has updated the project description as per template version 4.3 and requirement of VCS standards version 4.7 and it is found inline and consistent thus accepted. **Thus, CAR is closed.**	

CAR ID	02	Section no.	3.3	**Date:** 01-April-2024
Description of CAR				
1. Tool 27-version 12 'Investment analysis' is mentioned under the Section 3.1 and Section 3.2 of the PD version 01. Kindly update the Tool 27 – 'Investment analysis' as per the latest Tool 27, version 13.0 in the MR. 2. Under section 5.1 of the PD version 01, Data / Parameter- $EF_{Grid,CM,y}$ Justification of choice of data or description of measurement methods and procedures applied is found missing. 3. Under section 5.2 of the PD version 01, Data / Parameter-$EG_{PJ,y}$ 'Description of measurement methods and procedures to be applied' is found missing. **Thus, corrective action sought.**				
Project participant response				**Date:** 08-July-2024
1. Kindly refer to revised VCS PD for the latest version of Tool 27 (version 14.0) mentioned throughout the VCS PD 2. Kindly refer to revised VCS PD for the updated Data / Parameter- $EF_{Grid,CM,y}$ Justification of choice of data or description of measurement methods and procedures applied under section 5.1 3. Kindly refer to revised VCS PD for the updated Data / Parameter - $EG_{PJ,y}$ 'Description of measurement methods and procedures to be applied' under section 5.2				
Documentation provided by project participant				
Revised VCS PD				
VVB assessment				**Date:** 22-July-2024
1. Verifier found that the latest version of CDM tool 27 (tool for investment analysis) i.e. version 14 has been referred in the revised VCS PD and has been made consistent throughout the PD.				

2. PP has revised the section 5.1 in the PD and the details regarding "data / parameter- $EF_{Grid,CM,y,}$ justification of choice of data or description of measurement methods and procedures applied" have been provided and are consistent to project activity. 3. PP has revised the section 5.2 in the PD and the details regarding "data / parameter- $EG_{PJ,y}$ (or $EG_{facility,y}$) description of measurement methods and procedures to be applied" have been provided and are consistent to project activity. **Hence, CAR is closed.**

APPENDIX 4: COMPETENCE OF TEAM MEMBER AND TECHNICAL REVIEWER

Validation Team Member: -

No.	Role	Type of resource	Last name	First name	Affiliation (e.g. name of central or other office of VVB or outsourced entity)	Involvement in			
						Desk review	On-site inspection	Interview(s)	Validation findings
1.	Lead Auditor/ Technical Expert	OR	Shrivastava	Mr. Ishan	TQC-Outsourced entity	Yes	Yes	Yes	Yes
2.	Financial Expert	OR	Takarkhede	Dr. Atul	TQC-Outsourced entity	Yes	No	No	Yes
3.	Auditor/Technical Expert in Trainee	OR	Sharma	Mr. Khagesh	TQC-Outsourced entity	Yes	Yes	Yes	Yes
4.	Auditor-in-Training	OR	Singh	Ms. Anjali	TQC-Outsourced entity	Yes	Yes	Yes	Yes
5.	Observer	OR	Jain	Mr. Sarthak	TQC-Outsourced entity	Yes	Yes	Yes	Yes
4.	Observer	OR	Jangid	Mr. Ravi	TQC-Outsourced entity	Yes	Yes	Yes	Yes

Technical Reviewer and Approver of the Validation Report: -

No.	Role	Type of resource	Last name	First name	Affiliation (e.g., name of central or other office of VVB or outsourced entity)
1.	Technical reviewer (TR)	IR	Singh	Dr. N. Premjit	Applus+ Certification
2.	Approver	IR	Calle de Miguel	Agustin	Applus+ Certification

Shorts CV of the Team:

1. **Dr. Atul Takarkhede** is Ph.D. (Environmental Sciences) from Institute of Science, RTM Nagpur University, Nagpur, and he has already published different technical papers related to environmental sciences. He counts with more than 11 years of experience in field of Environmental Auditing, consulting and accreditation. He is an expert in ISO 9001-14001, CO2/GHG Reporting, Carbon Foot Print, Energy, Water and Waste Management reporting for organization environmental performance. His professional portfolio is mainly related with carrying out EIA, conducting QA/QC of EIA Reports; conducting environmental/water audits; NABET requirements appliance, functional area expert in Water Pollution & Solid & Hazardous Waste management among others. Furthermore, he counts with solid experience on CDM-VCS-GS consultancy and auditing. Currently he is associated with True Quality Certifications Private Limited and empanelled with Applus+ Certification to carry out GHG audits in the aforementioned schemes. Dr. Atul Takarkhede is based in Nagpur, India. Dr. Atul Takarkhede participates as part of the Audit Team as the Financial Expert for the assessment.

2. **Mr. Ishan Shrivastava**, has done bachelor of engineering in Mechanical Engineering from Rajiv Gandhi Proudyogiki Vishwavidyalaya, India. He has a year of working experience in India's one of the Maharatna Company i.e. GAIL (India) Limited in the area of Natural Gas, Energy & Environment. Currently. He is working in True Quality Certifications Pvt. Ltd. (An outsource entity for LGAI Technological Center, S.A. (Spain) "Applus+ Certification") since 2019 and has been involved as Auditor (Validator/Verifier) for Validation and Verifications of Project Activities (Renewable and non-renewable projects) under CDM/VCS/GS4GG/GCC programs. Mr. Ishan Shrivastava is based in Indore, India. Mr. Ishan Shrivastava participate as part of the Audit Team as Lead Auditor and Technical Expert for the assessment.

3. **Mr. Khagesh Sharma** has done bachelor of engineering in Electronic and Communication Engineering from Rajiv Gandhi Proudyogiki Vishwavidyalaya (Bhopal), India. He has 3 years of experience as Engineer, involve in the solar project design & consulting, Operation and maintenance activities. Currently, he is working in True Quality Certifications Pvt. Ltd. (An outsource entity for LGAI Technological Center, S.A. (Spain) "Applus+ Certification") since 2022 and has been involved in supporting Audit teams for Verifications of Project Activities (Renewable and non-Renewable projects) under CDM/VCS/GS4GG programs.

4. **Ms. Anjali Singh** holds a Bachelor of Engineering degree in Electronic and Communication Engineering from Rajiv Gandhi Proudyogiki Vishwavidyalaya, India. She has one year of experience as a Graduate Engineer Trainee at Idea Cellular Limited, India. She has also worked for three years as an Assistant Engineer at Gargee Energies, India, where she was involved in solar rooftop project design and consulting, as well as operation and maintenance activities. Currently, Anjali is working at True Quality Certifications Pvt. Ltd. She has been working there since 2021 and the company is an outsource entity for LGAI Technological Center, S.A. (Spain) "Applus+ Certification". At True Quality Certifications, Anjali supports audit teams for verifications of project activities for renewable and non-renewable projects under the Clean Development Mechanism (CDM), Verified Carbon Standard (VCS), and Gold Standard for Global Goals (GS4GG) programs. She is qualified Technical Expert in Scetorial Scope 1.2 (Energy Industries (Renewable / Non-Renewable Sources). She participates as Auditor-in-Trainee (Validation) of project activity.

5. **Mr. Sarthak Jain,** has done Bachelor of Technology in Information Technology Engineering from Walchand Institute of Technology, Solapur, Maharashtra, India affiliated with Solapur University. Currently, He is working in True Quality Certifications Pvt. Ltd. and has been involved in supporting Audit teams for Verifications of Project Activities (Renewable and non-Renewable projects) under CDM/VCS/GS4GG/GCC programs. Mr. Sarthak Jain is based in Indore (Madhya Pradesh), India. He particiapte as Observer in validation of project activity.

6. **Ravi Jangid Mr.** holds (Master's Degree in Environmental Engineering, Bachelor's Degree in Civil Engineering). His research pursuits and expertise include working on water, wastewater, and industrial wastewater performance evaluation and optimization with MATLAB on WTP/STP/CETP. He also interned with the World Resource Institute (WRI) on the Clean air catalyst project, which includes a source apportionment assessment of Indore city and different regulatory studies on waste burning, transportation, eateries, brick kilns, and so on. He furthermore completed research-based training at CPCB, where he has undertaken water, wastewater, and SWM testing. In addition, he has attended conferences, training programs, webinars, and gained certification in areas such as SWM, water, wastewater testing, air pollution monitoring. He also had a review paper published in the journal listed in UGC care list. He has been with True Quality Certifications Pvt. Ltd. (An outsource entity for LGAI Technological Center, S.A. (Spain) "Applus+ Certification") since 2023 and has been involved in supporting Audit teams for Validation/Verifications of Project Activities (Renewable and non-Renewable projects) under CDM/VCS/GS4GG/GCC programs.He particiapte as Observer in validation of project activity.

7. **Dr. N Premjit Singh** has a PhD in Mechanical Engineering (Thesis: Design and development of a square parabolic dish system with a concentrated photovoltaic (CPV) module for performance improvement) from the Indian Institute of Technology (IIT) Madras, Chennai, India, awarded in 2021. M.Tech in Energy Technology, Tezpur University, Napaam, India (2007), and B.Tech in Mechanical Engineering (2005), NERIST, Nirjuli, India. He has extensive experience of about 7 years with DOEs, including UNFCCC CDM and other carbon related schemes (e.g., VCS, GS, GCC), and 5 years + in research projects, renewable energy, and energy audits. In Applus+ since March 2023, he has been the Product Assurance Manager for CDM/VCS/GS4GG/GCC Department to ensure the quality of the performance of different assessments. Coordinate the global team for technical reviews, and identify the training needs for the auditors and technical reviewers to improve the quality of reports. Holds experience as a Lead Auditor, Validator and Verifier for GHG mitigation projects and programmes of activities in Sectoral Scope 1.2 (Renewables), and is qualified as per Applus+ procedures. Dr. N Premjit Singh is based in Gurugram, India. Dr. N Premjit Singh participate as the Technical Expert in the project activity.

APPENDIX 5: ABBREVIATIONS

Abbreviations	Full texts
CAR	Corrective Action Request
CDM	Clean Development Mechanism
CEA	Central Electricity Authority
CL	Clarification request
CO_2	Carbon dioxide
CO_2e	Carbon dioxide equivalent
CPI	Consumer Price Index
DOE	Designated Operational Entity
DR	Document Review
EF	Emission Factor
EIA	Environmental Impact Assessment
ER	Emission Reductions
ERC	Energy Regulatory Commission
FAR	Forward Action Request
GHG	Greenhouse gas(es)
IMF	International Monetary fund
OM	Operating Margin
GWP	Global Warming potential
RBI	Reserve Bank of India
VVB	Validation/Verification Body

APPENDIX 6: CALIBRATION DETAILS

Sl.No.	Investor Name	Energy Meter Serial Number	Energy Meter Make	Accuracy Class	Meter Type / Model	Meter Testing Date	Calibration valid upto
1	Main Meter's	Q0944018	SECURE	0.2S	Apex 100	31-January-2024	30-January-2029
2		Q0944019	SECURE	0.2S	ELITE 440	31-January-2024	30-January-2029
3		Q0944020	SECURE	0.2S	Apex 100	31-January-2024	30-January-2029
4		Q0944021	SECURE	0.2S	Apex 100	31-January-2024	30-January-2029
5	Check meter's	Q0944022	SECURE	0.2S	Apex 100	31-January-2024	30-January-2029
6		Q0944023	SECURE	0.2S	Apex 100	31-January-2024	30-January-2029
7		Q0944024	SECURE	0.2S	Apex 100	31-January-2024	30-January-2029
8		Q0944025	SECURE	0.2S	Apex 100	31-January-2024	30-January-2029